

END-OF-STUDY PROJECT REPORT

Submitted in Partial Fulfillment of the Requirements for the
BACHELOR DEGREE IN COMPUTER SCIENCE

Field of Study : Software Engineering and Information Systems

Study and implementation of a Tunisian stock market prediction approach using Machine Learning

By
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ALFAKIH ASSIL

Conducted within NA Soft



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Dedications

This work is for my loving family and friends who have supported me throughout my life due to their continued belief and unceasing strength. So, to my family, I appreciate the patience and love that you always had for me. Thank you for being there every step of the journey.

I have been uplifted when in doubt by the friendship and faith in my capabilities from you, my dear friends. I hold your friendship very dear to my heart. This is proof of how much support and inspiration I have gotten collectively from everyone of you. Your being in my life means so huge, and I always appreciate it.

Assil Al Fakih

To my parents, whose love has been my foundation in all moments of achievement and adversity. To my comrades, who brought joy, togetherness and moments of shared happiness into my life.

To my instructors and advisers, for illuminating the way and nurturing my interest in academics. To all the difficulties encountered - they helped mold my personality by giving out lessons worth more than wealth. To sleepless nights, with each minute dedicated being a step towards transformation.

They have proven that with determination and endurance, dreams can become real as they have achieved different goals. However, setbacks have only made me to be more determined and strong. Those created memories are something everyone should keep throughout his or her life. May tomorrow be characterized with ultimate courage in exploring new things it offers. Thanks a lot for your contribution.

Idriss Chefai

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We want to thank many people who immensely supported and directed us through our project. Ms. Imen Hamrouni's continued mentorship and insightful feedback really enlightened us and made this project possible; she was always there for us no matter what issue we faced she always encouraged us to keep moving forward and look ahead.

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Lastly, we would like to extend our appreciation to both people who helped directly and indirectly to the creation of this project, no matter how small they contributed it played a role in the shaping of this project.

Contents

Dedication	i
Acknowledgment	ii
General Introduction	1
1 Project Context	3
1.1 Introduction	3
1.2 Presentation of the Host Company	3
1.2.1 Host Company	3
1.2.2 Field of activities	3
1.3 Project presentation	4
1.3.1 Project framework	4
1.3.2 Problem	4
1.3.3 Study and Criticism of Existing Solutions	4
1.3.4 Proposed Solution	6
1.4 Project Management	9
1.4.1 CRISP DM	9
1.4.2 Metrics	10
1.5 Modeling Language	11
1.6 Conclusion	11
2 State of the Art	12
2.1 Introduction	12
2.2 Introduction to Finance and Stock Market	12
2.2.1 Foundational Concepts	12
2.2.2 Market Dynamics and Analysis	13
2.2.3 Types of Market Analysis	14
2.3 Fundamentals of Predictive Modeling	15
2.3.1 Overview of Predictive Models	15
2.4 Commonly Used Algorithms and Techniques	17
2.4.1 Statistical Models	18
2.4.2 Machine Learning models	18
2.4.3 Deep Learning models	18
2.5 Detailed Examination of Utilized Architectures	19
2.5.1 Linear Regression and Arima Model	19
2.5.2 Random Forest	21
2.5.3 LSTM Networks	22
2.6 Conclusion	23

3	System Requirements and Modeling	24
3.1	Introduction	24
3.2	System Requirements	24
3.2.1	Requirement Analysis	24
3.2.2	Global use case diagram	25
3.2.3	Refinement of the Use Cases	25
3.3	System Modeling	28
3.3.1	Global class diagram	28
3.3.2	Sequence diagrams of our application	28
3.4	Conclusion	30
4	Data Analysis and prediction	31
4.1	Introduction	31
4.2	Data Preparation	31
4.2.1	Data Collection	31
4.2.2	Data Cleaning	33
4.3	Data Analysis	34
4.3.1	Descriptive Statistics	34
4.3.2	Correlation Analysis	35
4.3.3	Time Series Analysis	35
4.3.4	Distribution Analysis	36
4.3.5	Outlier Detection	37
4.3.6	Moving Average Analysis	38
4.3.7	Volatility Analysis	38
4.3.8	Seasonality Analysis	39
4.3.9	Relative Strength Index (RSI)	39
4.3.10	Price-Volume Analysis	40
4.3.11	Interpretations and Conclusions	41
4.4	Model Implementation	41
4.4.1	Simple Linear Regression	41
4.4.2	Multivariate Linear Regression	42
4.4.3	ARIMA Model Implementation	44
4.4.4	Random Forest Model Implementation	46
4.4.5	LSTM Model Implementation (First Version)	48
4.4.6	LSTM Model Implementation (Second Version)	52
4.5	Model Comparison	54
4.5.1	Comparison Table	55
4.5.2	Observations	56
4.6	Conclusion	57
5	Realisation	58
5.1	Introduction	58
5.2	Work Environment	58
5.2.1	Introduction	58
5.2.2	Tools	58
5.2.3	Technologies Used	59
5.2.4	Libraries	60
5.3	Presentation	61
5.3.1	Clustering Interface	61

5.3.2	Correlation Interface	62
5.3.3	Single Stock Analytics Interface	62
5.3.4	Single Stock Prediction Interface	63
5.4	Conclusion	63
General Conclusion		64
	Netography	65

List of Figures

1.1	Methodoly used for predicting stock prices	8
1.2	CRISP-DM	9
1.3	Unified Modeling Language Representation	11
2.1	Linear Regression	19
2.2	Random Forest Model	21
2.3	An LSTM unit.	22
3.1	Admin	24
3.2	Global use case diagram	25
3.3	Global class diagram	28
3.4	Clustering Sequence Diagram	28
3.5	Correlation Sequence Diagram	29
3.6	Individual Stock Sequence Diagram	29
4.1	Data Content	33
4.2	Descriptive Statistics of BNA stocks	34
4.3	correlation of BNA Stocks	35
4.4	Time series Analysis of Bna Adjusted close price	35
4.5	Distribution Analysis of BNA Adjusted close price	36
4.6	Analysis of BNA volume	36
4.7	Outlier detection on BNA close prices	37
4.8	Outlier detection on BNA volume	37
4.9	Moving Average BNA	38
4.10	BNA volatility	38
4.11	Seasonality Analysis of the BNA	39
4.12	BNA stock RSI	40
4.13	Price-Volume Analysis of BNA	40
4.14	Plot actual and predicted values Linear Regression Version 1	42
4.15	Linear Regression v1 evaluation	42
4.16	Actual vs predicted close prices plot of Multivariate Linear regression	44
4.17	Arima model build summary	45
4.18	Arima pLot of predicted day	46
4.19	Random Forest Scatter plot close prices and actual prices	47
4.20	Random Forest future day prediction plot	48
4.21	Windowed Data	49
4.22	Training Predictions vs Training Observation	50
4.23	Validation Predictions vs Validation Observations	50
4.24	Test Predictions vs Test Observations	51

4.25	Prediction of close prices vs Observations	51
4.26	Prediction of close prices vs Observations & future close prices	52
4.27	Summary architecture LSTM v2	53
4.28	100 day prediction vs 100 day real close price	54
5.1	Clustering Interface	61
5.2	List of stocks	62
5.3	Correlation Interface	62
5.4	Single Stock Analytics Interface	63
5.5	Single Stock Prediction Interface	63

List of Tables

1.1	Comparison of Analytical Methods	5
1.2	Challenges in Stock Price Prediction Models	5
1.3	Applications of ML in Finance	6
2.1	Key Concepts in Finance	12
2.2	Factors Influencing Stock Prices	13
2.3	Methods of Stock Analysis	14
2.4	Statistical Models in Finance	18
2.5	Machine Learning Models in Finance	18
2.6	Deep Learning Models in Finance	18
3.1	Description of use case "Consult Clustering"	26
3.2	Refinement of the use case "Consult Individual Stock"	26
3.3	Refinement of the use case "Consult Interactive Analysis"	26
3.4	Refinement of the use case "Consult Prediction"	27
3.5	Refinement of the use case "Predict Stock"	27
3.6	Refinement of the use case "Consult Correlation"	27
4.1	Performance metrics for different models	55
4.2	Models and Descriptions	55
4.3	Models and Observations	56
5.1	Software Tools Description	58
5.2	Technologies Used	59
5.3	Technologies Used	60

General Introduction

Utilization of advanced methodologies in order to counter complex environments and strive to remain competitive is essential in today's changing environment concerning projects mainly within finance and technology. The purpose of this report is to guide us through an in depth understanding on how we can employ forecasting modeling methods in the context of financial analysis.

It is important to understand the importance of this project as we plunge into this venture in light of both financial and technological domains. The data-rich and more complex financial markets of today make the traditional analytical approaches insufficiently responsive to the fine market dynamics. Therefore, innovation is essential to provide predictive insights that can inform policy decisions..

Our goal contains these main expectations: first, to embrace potential predictive modeling via use of machine learning tools; and second presenting these results within an intuitive web application bespoke for financial insights. This serves as a foundation for our audience, pointing out the direction of the discussion and highlight the significance of our project in addressing present-day challenges. Then, we use the distinct chapters of the document to explore crucial aspects of our effort:

- **Chapter 1 : Project Context**

In this section, an explanation is given introducing the host, giving an idea scope of what our project is all about. We scrutinize current solutions that have not worked out so well and proffer new way forward. Similarly we detail how project will be managed including measures that track development.

- **Chapter 2 : State of the Art**

Let us journey through the vast world of prediction machines, explaining basic ideas and looking at popular algorithms and methods. Also, connecting finance and stock market, understanding underlying principles together with different ways of analyzing it.

- **Chapter 3 : System Requirements and Modeling**

The focus of this chapter is to delineate the system requirements and architectural modeling that will be used in the mentioned project. In this chapter, we are going to start by analyzing the requirements before proceeding to system modeling which will provide an insight on the structuring bases required for our an indication methods work efficiently.

- **Chapter 4 : Data Analysis and Prediction**

This is one of the critical phases of our work in which we take a closer look at

the manipulation, analysis or implementation of different datasets. From gathering data to discovering exceptions, many analytical methods are used in order to obtain valuable information. Also, we look into numerous forecast models, giving a full assessment of how well they work.

- **Chapter 5 : Realization**

Finally, we end our conversation by explaining the practical realization of our project. From explaining the work environment to showing tools and technologies used, we give a comprehensive view of the implementation stage.

In the end, this report is aim is to discuss the complicated nature of predictive modeling in finance through both theoretical arguments as well as real-world case studies that can help decision makers make better choices regarding the current market environment.

Using our methodical strategy, we hope to offer readers a thorough grasp of our project, from the initial idea to its implementation, and how it impacts the wider realm of financial analysis and forecasting.

Chapter 1

Project Context

1.1 Introduction

This chapter discusses the use of machine learning(ML) for financial forecasting in relation to NA Soft's activities in Tunisia's financial market. We examine current solutions, suggest fresh perspectives and describe project management using the CRISP-DM methodology. As well as why Unified Modeling Language (UML) is important in software development. This latter chapter provides a brief project overviews on its aims, methods, as well as implications in terms of how it can be employed specifically to enhance ability in financial planning with machines or computer systems doing parts that human beings cannot handle alone.

1.2 Presentation of the Host Company

To better understand the environment in which this project will be implemented, we'll first take a look at the host company.

1.2.1 Host Company

NA Soft is a leading provider of online trading services for the majority of stock exchange brokers in Tunisia, exceeding 55% of the local market share. At the heart of the financial market we do lies the responsibility and the opportunity to build a sustainable, equitable and more secure future for fintech applications.

1.2.2 Field of activities

Understanding the diverse activities of NA Soft will provide insight into the scope and potential impact of the project.

- **Electronic Document Management** : This includes organizing and managing documents digitally to make it simple to store, recover, and administer important electronic information. This facilitates operations such as sharing out documents, collaboration on projects among different workers, and archiving them, replacing physical paperwork with electronic systems while improving efficiency.

- **eHealth Application :** These are digital tools and platforms created to aid in the provision of quality healthcare while helping to improve the quality of patient care. Their different functions can be appointment scheduling, electronic health records (EHR) management, telemedicine consultations, medication tracking and health monitoring. eHealth apps are for making healthcare more accessible to people who need it – making it both cheap & good with convenience.
- **Financial Web Sites Development :** This means that developing websites or online platforms that are exclusively designed for finance such as banking, investment, insurance or trading. Account management systems, as well as transaction processors and financial plan tools among others, are some of the features provided by this kind of websites. The aim of these websites is to offer a secure and convenient route for clients' access and control over their own finances.
- **Stock Exchange Applications :** Software applications are these that are built for investors, merchants and financial experts such as stock market traders with the aim of keeping track with or buying selling stocks or shares as well as different kinds of financial instruments. Real times updates of news in forex markets, stock exchange are found which helps those in technical analysis. Trading platforms available give provision for execution while at the same time enabling people make informed buying or selling. In other terms they also allow individuals take part effectively within the fiscal domain.

1.3 Project presentation

In this part we are going to talk about the Project Framework, the problem encountered, then we are going to mention the challenges regarding stock price prediction, defining the solution we chose and lastly we are going to explain the management of our project and the modeling language we used.

1.3.1 Project framework

Our main goal is to use ML to predict future stock prices through a machine-learning driven website that is very straightforward. This project is of fundamental importance in the finance and technology sectors since the markets are increasingly intricate and data is abundant.

1.3.2 Problem

We aim to make a ML model allowing us to predict the performance of the stock market. It should also help those working in finance markets make good decisions based on data. Traditional finance analysts may not always anticipate the future since their focus is on the historical data. Our aim is to fill that gap by connecting traditional finance analysis with ML, which can be more accurate and provide it on time.

1.3.3 Study and Criticism of Existing Solutions

Over the past few years, there have been different methods suggested for forecasting stock prices using conventional statistical models, ML algorithms, and quantitative techniques.

These strategies are designed to utilize past data, market signals and economic factors to predict future price changes thus guiding investment choices. Some popular methods are in the table 1.1 below:

Table 1.1: Comparison of Analytical Methods

Method	Definition / Strengths and Weaknesses
Time Series Analysis	Analyzing time series by statistical means refers to observing trends in data as they unfold over time. + Provides accurate predictions for future trends – Assumes historical patterns will continue in the future
Technical Analysis	Analyzes historical price and volume data to identify potential trading opportunities based on chart patterns and indicators + Useful for identifying short-term trading opportunities based on price patterns – Relies heavily on historical price data and may not account for fundamental factors
Fundamental Analysis	Evaluates the intrinsic value of a security by analyzing economic and financial factors, such as earnings, revenue, and market conditions + Offers insights into the intrinsic value of assets based on financial and economic data – Interpretation can be subjective and may not capture market sentiment or short-term price movements
ML Methods	Utilizes algorithms that learn from data to make predictions or decisions without being explicitly programmed + Can handle complex data patterns and adapt to changing market conditions – Requires substantial data preprocessing and tuning of models; interpretability can be a challenge
Sentiment Analysis	Analyzes sentiment from textual data (e.g., social media posts, news articles) to gauge public opinion for decision-making + Analyzes sentiment from textual data for decision-making – Performance can be affected by nuances in language and context; requires high-quality data and language processing techniques

Challenges in Existing Stock Price Prediction Models

In the following table 1.2 are some criticisms of existing solutions for predicting stock prices, despite the valuable insights and tools they provide to investors.

Table 1.2: Challenges in Stock Price Prediction Models

Challenge	Description
Limited Predictive Power	Accurate stock price prediction is difficult for a variety of models, particularly in active markets. Sometimes, especially when they are not able to capture intricate connections, normal models produce predictions that cannot be trusted.

Continued on next page

Table 1.2 – Continued from previous page

Challenge	Description
Data Quality Issues	Stock prediction models, as is the case with artificial intelligence, are greatly influenced by low quality data such as information omissions and prejudices hence rendering the ability to build firm, trustworthy models impossible.
Overfitting and Model Complexity	Overfitting will happen when you train a model with data that it fits very well, but the moment it comes across new evidence or information, that model starts performing poorly. In particular, this phenomenon is common among more advanced models such as deep neural networks that have been trained using little data containing noise.
Lack of Interpretability	Investors are hesitant to trust and take up ML models of stock prediction, for example, neural networks and ensembles, due to their complexity and interpretability.
Market Efficiency Hypothesis	According to the efficient market hypothesis stock prices reflect all available information making it difficult for consistent market outperformance with the use of prediction models because they are too slow in predicting changes in prices.

1.3.4 Proposed Solution

The idea is to use ML algorithms to forecast stock prices more accurately and reliably. By using advanced data analytics and predictive modeling, the goal is to give investors helpful information on market trends, find profitable trading chances, and improve investment strategies.

Role of ML in Finance

ML has become a valuable tool in the financial industry, allowing professionals to analyze data, predict trends, and make informed investment decisions. By using advanced computational techniques, ML algorithms can uncover patterns, relationships, and anomalies within large datasets, helping financial experts extract valuable insights and achieve excess returns in the market. The applications of ML in finance span across various key areas, including these in the table 1.3:

Table 1.3: Applications of ML in Finance

Application	Description
Predictive Analytics	Regression analysis, time series forecasting, and classification are some of the AI algorithms that can be used to predict market changes, asset values, and economic indicators. Past price data, financial metrics, and market conditions analysis improve stock price predictions by ML.
Risk Management	AI algorithms find out and handle investment dangers through the assessment of the portfolio volatility, value-at-risk, and abnormalities in financial data, detecting the irregularities and fraudulent behavior, thus helping in the regulatory compliance and minimizing the losses risk.

Continued on next page

Table 1.3 – Continued from previous page

Application	Description
Algorithmic Trading	ML algorithms are used to automate the process of trading based on a set criteria as well as signals from the market. High-frequency trading firms take advantage of market inefficiencies and execute transactions quickly enough using artificial intelligence technology.
Portfolio Management	ML algorithms help improve investment options and ways of distributing them across assets. This they achieve by looking at the historical market data, and investor preferences as well risk factors. They make use of mean-variance optimization plus Monte Carlo simulation technique in order to increase management of risk hence maximizing returns.
Sentiment Analysis and Market Forecasting	The ML process studies raw data (i.e. news articles, social media posts) in order to gauge market sentiment and estimate shifts in the attitudes of investors. Essential market topics and the direction of human feelings about them are identified through natural language processing (NLP) methodologies.

Artificial intelligence has been making a huge impact on the finance industry. It offers advanced tools for analyzing financial data, predicting future trends, and guiding investment choices. Using AI algorithms, financial experts can better understand market behaviors, create successful trading plans, and improve portfolio management strategies. This leads to improved investment results and benefits for investors.

Proposed solution Benefits

The recommended solution provides numerous advantages for investors, traders, and financial institutions:

- **Enhanced Predictive Accuracy**

Using sophisticated ML algorithms and predictive modeling techniques, the solution is designed to enhance the precision and dependability of stock price forecasts in comparison to conventional statistical models. By identifying intricate patterns and connections within the data, investors are offered more precise predictions of future price fluctuations.

- **Timely Insights and Decision Support**

Investors can use the proposed solution to stay updated on market trends, spot new investment options, and make smart decisions. The solution uses real-time market data and economic indicators to offer tools that assist investors in refining their trading strategies and portfolio distributions.

- **Risk Management and Portfolio Optimization**

The solution provides tools for managing risk and optimizing portfolios to help investors effectively diversify holdings and increase returns. By considering risk factors, estimating volatility, and conducting correlation analysis, the solution empowers investors to create robust portfolios that perform effectively in varied market environments.

- **Scalability and Flexibility**

The suggested solution is very adaptable and versatile, making it able to adjust

to changing market trends and investor choices. By using cloud computing and scalable data processing methods, the solution can rapidly and effectively analyze large amounts of data, making it appropriate for both individual investors and institutional clients.

- **Transparency and Interpretability**

Compared to black-box models, the suggested solution focuses on being transparent and easy to understand, allowing investors to grasp the process behind predictions and confirm the accuracy of model results. Through offering clear feature importance scores, model evaluations, and visual aids, the solution boosts credibility and assurance in the accuracy of predictive models.

Methodology and Approach

Our solution takes a holistic approach to predicting stock prices by combining various ML algorithms and techniques to analyze past data, identify significant patterns, and make educated guesses about future price changes. The process includes the following essential steps in the figure 1.1:

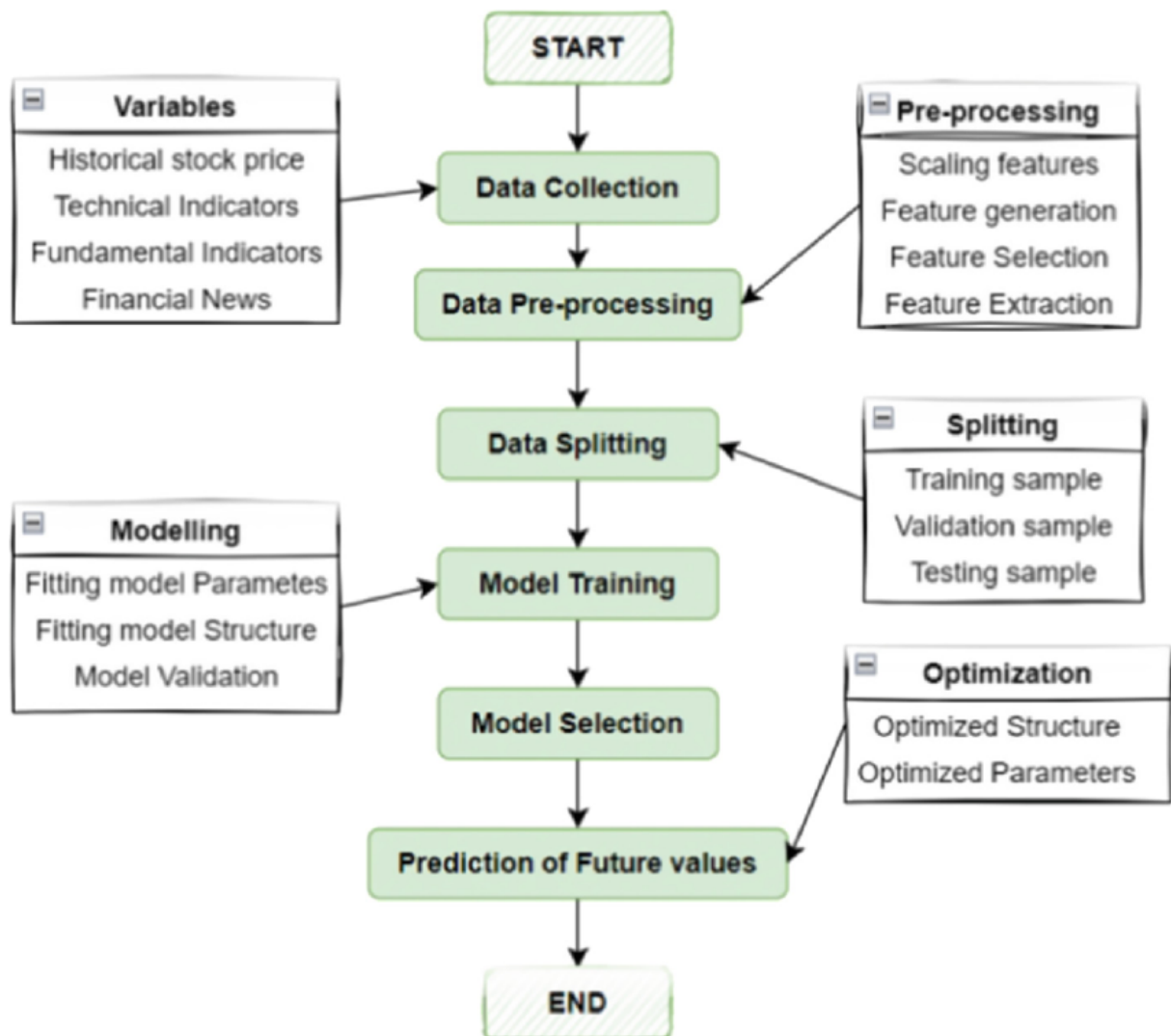


Figure 1.1: Methodoly used for predicting stock prices

1.4 Project Management

The CRISP-DM methodology provides a structured approach to data science projects. Let's explore its phases and relevance to our project.

1.4.1 CRISP DM

In industry, they use standard data mining methodologies called CRISP-DM or Cross-Industry Standard Process for Data Mining. This is a proven and tested good approach. A methodology which is recommended by 45% of people who participated in survey dated 2014 aimed at data scientists and analysts. CRISPDM comprises of six process, each with unique deliverables.

- **Understanding the Business Context:** Understanding the problem from the business domain is crucial before getting into technological solutions.
- **Data Understanding:** Identify and familiarize with the data that is available or find out what data should be collected.
- **Data Preparation:** This means that various tasks are undertaken including standardization of data, addressing lost values and ensuring that the data is clean.
- **Modeling:** Get the model ready and develop it from the processed data.
- **Evaluation:** Evaluate the performance of the model against predetermined metrics.
- **Deployment:** If the model meets requirements, deploy it into production; otherwise, revisit the initial stages for refinement.

Below is the figure 1.2 representing the CRISP DM methodology

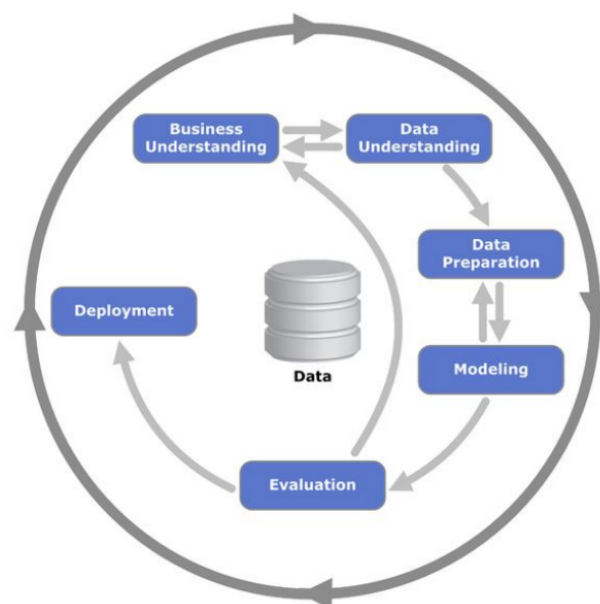


Figure 1.2: CRISP-DM

1.4.2 Metrics

As a way to compare models, we must define a specific metric that measures the efficiency of each model. We are going to dismiss any classification metric since the problem at hand is a regression problem. The performance of the model is calculated using the difference between the target and the predicted value.

Mean Absolute Error (MAE)

$$\text{MAE} = \frac{1}{n} \sum |y - \hat{y}| \quad (1.1)$$

Equation 1.1 is the average of the absolute values of the differences between the target and predicted values. Using absolute value is important; the sign of the error does not matter in having a clear metric, however its magnitude does. Note that $\hat{y}$ here represents the forecasted value.

Mean Absolute Percentage Error (MAPE)

$$\text{MAPE} = \frac{1}{n} \sum \left| \frac{y - \hat{y}}{y} \right| \quad (1.2)$$

Equation 1.2 calculates the average of percentages expressing the accuracy of the forecast. Using percentages is an intuitive way of expressing success over failure. However, using MAPE is not recommended: It penalizes negative values more than positive ones. It does not allow the target value to be zero (division by zero).

Mean Square Error (MSE)

$$\text{MSE} = \frac{1}{n} \sum (y - \hat{y})^2 \quad (1.3)$$

Equation 1.3 is the average of the squared differences between the target value and the predicted value; squaring the difference penalizes the model for larger errors and makes negative difference positive. This makes MSE more sensitive to outliers than the other metrics we mentioned.

Root Mean Square Error (RMSE)

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum (y - \hat{y})^2} \quad (1.4)$$

Equation 1.4 is the square root of MSE; calculating its square root makes the metric more intuitive as the values would be much smaller. RMSE loss value also shares the same unit as the original values.

Root Mean Square Error gives a clear evaluation of the performance of the model. It is not biased (unlike MAPE). For these reasons, we are using it as a metric.

1.5 Modeling Language

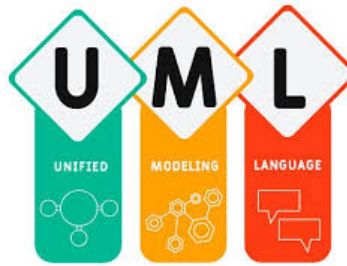


Figure 1.3: Unified Modeling Language Representation

The figure 1.3 represent the Unified Modeling Language (UML) and it is an established system for viewing and building software systems. Using this software, there is an opportunity to come up with efficient, flexible as well as secure software components. In object-oriented systems development, Unified Modeling Language is used as it helps in modeling software system graphically using notations for objects. The principles of Object Management Group form the basis for its operation.

Two principal types of UML diagrams can be identified: those illustrating structure and those showing behavior. They deliver a coherent perspective of the software system logics such as one that demonstrates the kinds of classes in the program, their attributes, as well as connections between them- Class Diagram.

On the other hand, behavioral diagrams detail how software systems work. Use case diagram is an example that shows the system features in terms actors, goals, and dependencies, while sequence diagram shows communication between system components across different messages.

1.6 Conclusion

In Conclusion, this chapter contain researches on the usage of ML in financial forecasting within NA Soft's operations in Tunisia's financial market. Criticisms have been made of current approaches, some new ways suggested, and a comprehensive project plan using the CRISP-DM method detailed. In addition, we discussed some important numbers that are used to measure how well models work, and emphasized why Unified Modeling Language (UML) is important for creating software. During these discussions, we examined how ML could be used to help improve financial forecasting by providing information that investors and analysts can make use of. This section is used to lay a basis for other discussions, initiating more examination of the use and assessment of algorithms for ML regarding financial prediction.

Chapter 2

State of the Art

2.1 Introduction

Chapter 2 is devoted to exploring, firstly finance and stock markets. And secondly advanced predictive modeling architectures that are employed in a number of areas such as speech synthesis, sequence prediction, and financial modeling. From linear regression and ensemble methods such as Arima and Random Forest, to the mighty Long Short-Term Memory (LSTM). This is a range through which various modern techniques are presented. These patterns function as the bedrock that enables one to handle difficult forecasting chores while boosting the quality of judgments made..

2.2 Introduction to Finance and Stock Market

This section includes a detailed explanation of what finance entails as well as stock markets by looking into basic principles, operations within them among other things like how different aspects such as trends shape them while assessing ways of studying them. To make well-informed choices when venturing into the present day's financial world, one must know these basics to heart.

2.2.1 Foundational Concepts

Finance is a complex field that deals with handling money, investments, and assets. It involves various tasks such as financial planning, budgeting, investing, and managing risk. Ultimately, the goal of finance is to efficiently allocate resources to meet economic goals and create value for stakeholders. In the following table (2.1) are some terms related to the stock market.

Table 2.1: Key Concepts in Finance

Concept	Description
Stocks and Stock Markets	Stocks represent ownership in publicly traded companies. The stock market facilitates trading of stocks, allowing investors to buy and sell shares to determine their value.

Continued on next page

Table 2.1 – Continued from previous page

Concept	Description
Bonds	Bonds are debt securities used by governments or corporations to borrow money. Investors who buy bonds receive regular interest payments and the principal amount back when the bond matures.
ROI (Return on Investment)	ROI measures the profitability of an investment relative to its initial cost. It is calculated by dividing the net profit gained from the investment by the initial investment cost and expressing the result as a percentage.
Portfolio	A portfolio is a collection of investments owned by an individual or entity, consisting of diverse assets like stocks, bonds, real estate, and commodities. Portfolios are tailored based on financial objectives and risk tolerance.
Assets	Assets are valuable resources owned by individuals, businesses, or groups that can generate income or provide advantages. Examples include money, investments, property, and intellectual property.
S&P 500	The S&P 500 is an index representing 500 large companies listed on U.S. stock exchanges. It serves as a benchmark for assessing the overall performance of the economy.

2.2.2 Market Dynamics and Analysis

Investors rely on an interplay of factors that help understand stock price movements.

Factors Influencing Stock Prices

When it comes to stock prices, there are many factors at play. These can range from internal company matters to external market conditions. Investors and analysts need to have a good grasp of these elements in order to make wise investment choices. Some of the main factors that impact stock prices are in the table 2.2 below:

Table 2.2: Factors Influencing Stock Prices

Factor	Description
Company Performance	An investor's feelings about a company's future and the price that people are willing to pay for its stock can both be changed by variables such as increase in earnings, profit margins, etc. or factors affecting cash flow – including slow cash burn. To assess whether it is worth investing in or buying shares in organization A or choosing another alternative instead the investor must become acquainted more with its quarterly profit results arising from trading activities as well as other financial indicators disclosed there.
Market Sentiment and Investor Psychology	Fear, greed, and uncertainty are crucial emotions when it comes to stock price movements in the short term, according to investors. This is because market disruptions are often brought about by political conflicts as well as economic news.
Industry Trends and Competitive Positioning	Industry success is affected by technological advances, changes in regulation and consumer preferences. Companies outperform competitors because they have strong market positions and innovate with better products.

Continued on next page

Table 2.2 – Continued from previous page

Factor	Description
Macroeconomic Indicators	Investor sentiment and market dynamics are impacted by factors like interest rates, inflation, Gross Domestic Product (GDP) growth and unemployment rates. The central bank's decisions on monetary policy, in turn, affect borrowing costs as well as consumer spending patterns thus affecting stock prices in various segments of the economy.
Global Economic and Political Events	Global incidents just like exchange disputes and geopolitics effect markets internationally. Elections and government actions greatly impact how persons who have invested perceive themselves and how they think about the future of markets and this eventually stems to changes in stock prices.
Investor Behavior and Market Trends	As far as bull and bear markets and sector rotations are concerned, it is collective investor actions that drive the trends. Market trends are detected by technical analysis, which uses price momentum and chart patterns.
Currency Fluctuations and Exchange Rates	Multinational corporations are affected by currency fluctuations, especially if they are moving a lot. This in turn affects global trade dynamics and export opportunities which has further implications on the stock prices in interconnected markets.

It is important for investors and analysts to understand how these factors interact in order to evaluate the true value of stocks, spot investment chances, and navigate the ever-changing world of financial markets. Keeping up-to-date with economic indicators, industry trends, and market news allows investors to make well-informed choices and efficiently handle their investment portfolios.

2.2.3 Types of Market Analysis

Market analysis involves different methods and techniques used by analysts and investors to assess stocks and make well-informed investment choices. These methods can be broadly classified into three main types shown in the table 2.3 below:

Table 2.3: Methods of Stock Analysis

Method	Description
Fundamental Analysis	When investors examine a company's balance sheet, income statement, and cash flow statement along with subjective information (including competitiveness, trade trends, and management competence), they are able to determine whether the stocks will be undervalued or overvalued for long-term sale; that is, how much they are worth.

Continued on next page

Table 2.3 – Continued from previous page

Method	Description
Technical Analysis	Traders love this method because this technique assists them in taking short-term profits depending on the patterns observed in the market yesterday. In details we check on trends involving both price and volume indicators as well as chart patterns. As indicators, moving averages, RSI and MACD are used; while for chart patterns, one would employ support/resistance levels, trend lines or candlestick patterns.
Quantitative Analysis	Mathematical models, statistical techniques and machine learning algorithms (regression, neural networks, decision trees) are utilized by analysts to evaluate financial data for stock price predictions and to identify trading prospects. Systematic traders as well as hedge funds opt for quantitative analysis because it entails data-based tactics and risk control.

There are different ways to analyze the market, each with its own advantages and disadvantages. Investors often use a mix of these methods to get a better understanding of the market and make smart investment choices. By using fundamental, technical, and quantitative analysis, investors can create strong investment plans that match their financial goals and risk tolerance.

2.3 Fundamentals of Predictive Modeling

This section overviews The methods, machine learning algorithms, and deep learning architectures applied in predictive analysis.

2.3.1 Overview of Predictive Models

From diverse fields, such as in different sectors, decision-making is often prompted by the prognostication of future events based on prior events through models that show some predictability. They include a combination of statistical analysis methods with data mining technologies to encompass various relationships among different elements within a system so that processes may be more effectively performed as well as strategies more efficiently designed.

Statistical Models

Statistical models are mathematical explanations on how the variables in datasets relate with each other. These models are created following statistical principles or theories and are mainly used in making predictions or guesses about the population by looking at the data.

Examples of statistical Models :

- **Linear Regression :**
A mathematical model is applied to portray the connection between a continuous reliant variable and one or more free variables.
- **ARIMA (Auto Regressive Integrated Moving Average):**
"A time series model is used for anticipations relying on past value and trend info."

- **Logistic Regression :**

A statistical method utilized in distinguishing objects from one another by assigning to particular objects in a hierarchical classification.

Machine Learning Models

Artificial Intelligence acts like the human brain simulators, which do their work through what is referred to as machine learning. They have the ability of continually improving in their performance with increase in the exposure to more data.

Examples of Machine Learning Models :

- **Random Forest :**

A method is created using multiple decision trees known as ensemble learning, which would then be responsible for improving accuracy as well as robustness by making predictions based on these decisions made by different predictors.

- **Support Vector Machines (SVM):**

It is used for classification and regression tasks and is particularly good at dealing with high-dimensional spaces.

- **Neural Network :**

Deep learning models consist of layers of artificial neurons connected to each other, with the capability of learning from large datasets some complex patterns that are in them.

Deep learning Models

Deep learning models, which are based on the neural network architecture of the human brain, use complex structures to train machines to understand data and make decisions based on it, for example recognizing patterns and categorizing information.

Examples of Deep Learning Models :

- **Feedforward Neural Networks (FNNs)**

FNNs process information unidirectionally, making them effective for tasks like image recognition in autonomous driving.

- **Recurrent Neural Networks (RNNs)**

RNNs maintain memory through recurrent connections, crucial for analyzing sequential sensor data in autonomous driving. The computation in an RNN at time step t is expressed in the equation 2.1:

$$h_t = f(W_{hh} \cdot h_{t-1} + W_{xh} \cdot x_t + b_h) \quad y_t = W_{hy} \cdot h_t + b_y \quad (2.1)$$

- **LSTMs and Gated Recurrent Units (GRUs)**

LSTMs and GRUs address RNN limitations, crucial for understanding traffic patterns in autonomous driving. They incorporate specialized units that selectively retain or forget information over multiple time steps. The mathematical formulations for LSTMs and GRUs are represented in the 2.2 equations and the 2.3 equation:

LSTMs:

$$\begin{aligned}
f_t &= \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \\
i_t &= \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \\
o_t &= \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \\
c_t &= \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \\
\text{cell}_t &= f_t \odot \text{cell}_{t-1} + i_t \odot c_t \\
h_t &= o_t \odot \tanh(\text{cell}_t)
\end{aligned} \tag{2.2}$$

GRUs:

$$\begin{aligned}
r_t &= \sigma(W_r \cdot [h_{t-1}, x_t] + b_r) \\
z_t &= \sigma(W_z \cdot [h_{t-1}, x_t] + b_z) \\
\tilde{h}_t &= \tanh(W_h \cdot [r_t \odot h_{t-1}, x_t] + b_h) \\
h_t &= (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t
\end{aligned} \tag{2.3}$$

- **Convolutional Neural Networks (CNNs)**

CNNs excel in hierarchical feature extraction, vital for object detection in autonomous driving. The mathematical formulation for a basic operation in a convolutional layer is as follows in the 2.4 equation:

$$\text{Conv}(x, w)_{i,j} = \sum_m \sum_n x_{i+m, j+n} \cdot w_{m,n} \tag{2.4}$$

- **Autoencoders and Generative Adversarial Networks (GANs)**

Autoencoders focus on data representation, while GANs generate realistic data, both used for dimensionality reduction and data augmentation in autonomous driving.

- **Transformers**

Transformers capture contextual relationships effectively through self-attention mechanisms, applicable in processing sequential sensor data for AVs.

- **Hybrid Architectures**

Hybrid architectures combine models like CNNs, RNNs, transformers, autoencoders, and GANs for enhanced performance and versatility in autonomous driving perception systems. These models leverage the strengths of various architectures to address the complex challenges encountered in autonomous driving scenarios.

2.4 Commonly Used Algorithms and Techniques

Artificial intelligence is extremely important in finance for tasks such as predictive analytics, risk management, algorithmic trading, and portfolio management. There are multiple algorithms and techniques utilized based on the specific goals. Here is a list of commonly used algorithms and techniques in finance:

2.4.1 Statistical Models

Table 2.4 provides an overview of statistical models commonly used in finance. These models include regression models and time series analysis techniques, which are essential for understanding and predicting financial data.

Table 2.4: Statistical Models in Finance

Method	Description
Regression Models	Statisticians use regression analysis to examine relationships between a dependent variable (e.g., stock price) and independent variables (e.g., financial data). Linear regression models like Ordinary Least Squares (OLS) are common for continuous outcome prediction in finance.
Time Series Analysis	Analyzing data collected over time to uncover patterns, trends, and seasonal variations. Methods like ARIMA, exponential smoothing, and Fourier transforms are used to model and predict time series data in finance.

2.4.2 Machine Learning models

Table 2.5 highlights key machine learning models used in finance. It includes descriptions of neural networks and support vector machines (SVM), which are powerful tools for pattern recognition, sequence prediction, and risk modeling in the financial sector.

Table 2.5: Machine Learning Models in Finance

Method	Description
Neural Networks	Machine learning algorithms inspired by the human brain, consisting of interconnected nodes arranged in layers. Deep learning methods like CNNs and RNNs are used in finance for pattern recognition, sequence prediction, and risk modeling.
Support Vector Machines (SVM)	Supervised learning algorithms for classification and regression tasks, finding optimal hyperplanes to categorize data or make predictions in finance (e.g., stock price prediction, credit risk assessment).

2.4.3 Deep Learning models

Table 2.6 presents deep learning models used in finance, focusing on time series forecasting with Long Short-Term Memory (LSTM) networks. These specialized RNN architectures are designed to handle sequences with long-range dependencies, making them ideal for financial forecasting tasks.

Table 2.6: Deep Learning Models in Finance

Method	Description
Time Series Forecasting with LSTM	Specialized RNN architecture for analyzing sequences with long-range dependencies. Used in finance for forecasting time series data (e.g., stock prices, volatility) due to their ability to capture temporal trends and complex relationships.

When financial professionals use these algorithms and techniques, they can understand market trends better, create models to predict stock prices, improve trading strategies, and manage portfolio risks more efficiently. This ultimately leads to better investment results and adds value for investors.

2.5 Detailed Examination of Utilized Architectures

In this section, we dive into fundamentals to understand and predict relationships within data, emphasizing on numeric predictions and time series analysis.

2.5.1 Linear Regression and Arima Model

In this part we are explaining the linear regression and the arima model architectures

Linear Regression

Linear regression fits a linear model through a set of data points to estimate the relationship between a target outcome label and one or more feature variables in order to predict a numeric value. The output y (label) can be estimated as a straight line that can be visualized as:

$$y = \text{intercept} + c_i * x_i + \text{Error}$$

Where x_i are the input variables (features), and the parameters c_i , intercept, and Error are the regression coefficients, the constant offset, and the error respectively. The coefficients c_i can be interpreted as the increase in a dependent variable (y label) for a unit increase in the respective independent variable (x feature).

In the simple example below (figure 2.1), linear regression is used to estimate the stock price (the y label) based on the stock Adjusted Low price(the x feature).

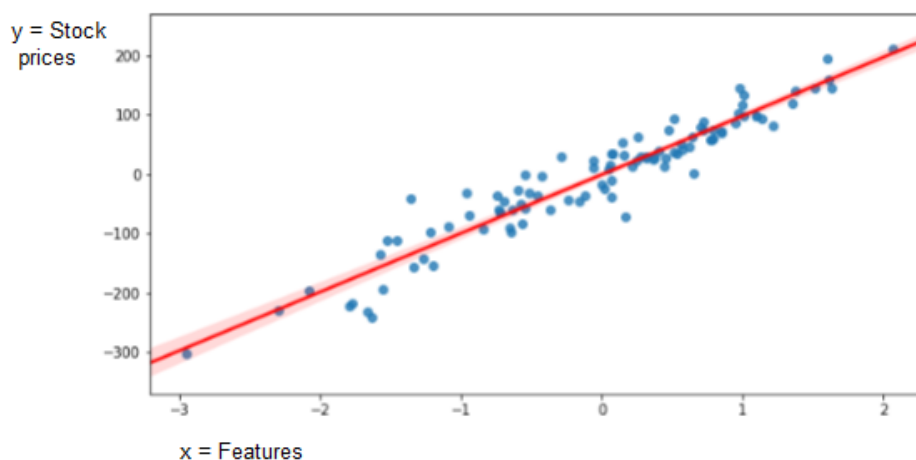


Figure 2.1: Linear Regression

The distance between the x and y points and the line determines the strength of the connection between the independent and dependent variables. The slope of the line is often determined using the least squares method in which the sum of the squares of the offsets of points on the curve is minimized.

Arima Model

The ARIMA (Autoregressive Integrated Moving Average) model is a statistical analysis model that uses time series data to either better understand the data set or to predict future trends.

$$X_t = \alpha + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \dots + \phi_p X_{t-p} + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q} + E_t \quad (2.5)$$

Where:

- X_t : Value of the time series at time t .
- α : Constant term.
- $\phi_1, \phi_2, \dots, \phi_p$: Autoregressive parameters of order p .
- $\theta_1, \theta_2, \dots, \theta_q$: Moving average parameters of order q .
- ϵ_t : Error term at time t .

A statistical model is autoregressive if it predicts future values based on past values. For example, an ARIMA model might seek to predict a stock's future prices based on its past performance or forecast a company's earnings based on past periods.

Model Components

The ARIMA(p, d, q) model consists of three components:

1. **Auto-Regressive (AR) Component:** The AR component models the relationship between the current value and past values. It's denoted by p , representing the number of lagged terms. For example, an AR(1) model can be written as in the 2.6 equation:

$$X_t = \phi_1 X_{t-1} + \varepsilon_t \quad (2.6)$$
2. **Differencing (I) Component:** The I component deals with differencing the data to make it stationary. It's denoted by d , indicating the number of times differencing is performed. It represents the differencing of raw observations to allow the time series to become stationary (i.e., data values are replaced by the difference between the data values and the previous values).
3. **Moving Average (MA) Component:** The MA component captures the relationship between the current value and past forecast errors. It incorporates the dependency between an observation and a residual error from a moving average model applied to lagged observations.

ARIMA Parameters

Each component in ARIMA functions as a parameter with a standard notation. For ARIMA models, a standard notation would be ARIMA with p , d , and q , where integer values substitute for the parameters to indicate the type of ARIMA model used. The parameters can be defined as:

- p : the number of lag observations in the model, also known as the lag order.
- d : the number of times the raw observations are differenced; also known as the degree of differencing.
- q : the size of the moving average (MA) window, also known as the order of the MA.

A value of zero (0), which can be used as a parameter, would mean that particular component should not be used in the model. This way, the ARIMA model can be constructed to perform the function of an ARMA model, or even simple AR, I, or MA models.

2.5.2 Random Forest

Ensemble learning methods combine multiple machine learning (ML) algorithms to obtain a better model—the wisdom of crowds applied to data science. They’re based on the concept that a group of people with limited knowledge about a problem domain can collectively arrive at a better solution than a single person with greater knowledge. Random forest is an ensemble of decision trees, a problem-solving metaphor that’s familiar to nearly everyone. Decision trees arrive at an answer by asking a series of true/false questions about elements in a data set. In the example below, to predict a person’s income, a decision looks at variables (features) such as whether the person has a job (yes or no) and whether the person owns a house.

In an algorithmic context, the machine continually searches for which feature allows the observations in a set to be split in such a way that the resulting groups are as different from each other as possible and the members of each distinct subgroup are as similar to each other as possible.

Random forest uses a technique called “bagging” to build full decision trees in parallel from random bootstrap samples of the data set and features. Whereas decision trees are based upon a fixed set of features, and often overfit, randomness is critical to the success of the forest.

Below in the figure 2.2 is an example of a random Forest Model

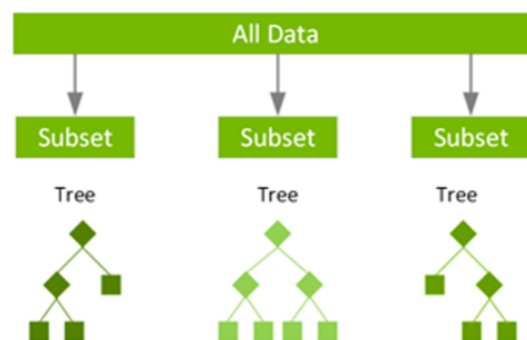


Figure 2.2: Random Forest Model

2.5.3 LSTM Networks

LSTM is a type of Recurrent Neural Network (RNN) specially designed to prevent the neural network output for a given input from either decaying or exploding as it cycles through the feedback loops.

The feedback loops are what allow recurrent networks to be better at pattern recognition than other neural networks. Memory of past input is critical for solving sequence learning tasks and LSTMs provide better performance compared to other RNN architectures by alleviating what is called the vanishing gradient problem. LSTMs due to their ability to learn long term dependencies are applicable to a number of sequence learning problems including language modeling and translation, acoustic modeling of speech, speech synthesis, speech recognition, audio and video data analysis, handwriting recognition and generation, sequence prediction, and protein secondary structure prediction.

LSTM Architecture

The LSTM Architecture consists of linear units with a self-connection having a constant weight of 1.0. This allows a value (forward pass) or gradient (backward pass) that flows into this self-recurrent unit to be preserved and subsequently retrieved at the required time step. With the unit multiplier, the output or error of the previous time step is the same as the output for the next time step.

This self-recurrent unit, the memory cell, is capable of storing information which lies dozen of time-steps in the past. This is very powerful for many tasks. For example for text data, an LSTM unit can store information contained in the previous paragraph and apply this information to a sentence in the current paragraph.

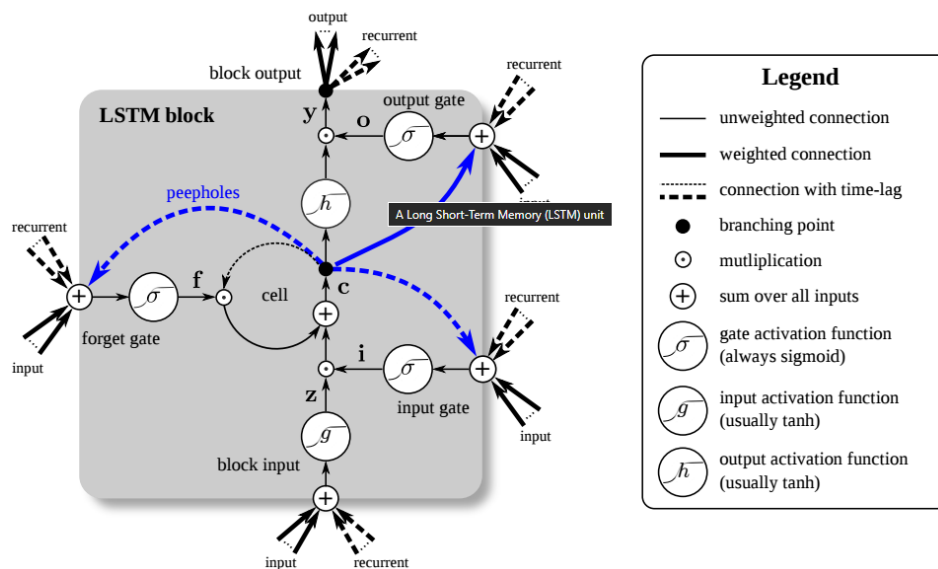


Figure 2.3: An LSTM unit.

Bidirectional LSTMs train the input sequence on two LSTMs - one on the regular input sequence and the other on the reversed input sequence. This can improve LSTM network performance by allowing future data to provide context for past data in a time series. These LSTM networks can better address complex sequence learning/ machine learning problems than simple feed-forward networks.

2.6 Conclusion

A detailed examination of the modern predictive architectures is given in this chapter. Integrating these methods with fundamental finance principles establishes a strong basis for solving investment management and portfolio optimization problems in actual practice. This holistic approach will give professionals important equipment & knowledge on how to use predictive modeling & finance in order to make informed decisions for value creation.

Chapter 3

System Requirements and Modeling

3.1 Introduction

In this chapter, we examine vital system requirements and modeling for our project. First, we start with understanding the system's needs as well as the global use case diagram and refinement. Second, essential aspects in system modeling will be discussed, hence giving a clearer picture about what this project aims to achieve.

3.2 System Requirements

In this section, we will elucidate the requirements of our system as well as system modeling.

3.2.1 Requirement Analysis

In this section, we will define who does what, as well as describe the functional and non-functional requirements that our project needs.

Actors

Our application will be used by one actor



Figure 3.1: Admin

Functional Requirements

Our project entails the development of a web application that offers predictive stock analysis. The following task will be performed by the Admin within the system:

- **Prediction:**

- The admin can select a stock for prediction
- The admin can input how many days he wishes to predict
- The admin can initiate the prediction process
- The admin can view the predicted stock prices and related data on a user-friendly interface

These functionalities facilitate various tasks such as:

- Predict future stock prices based on historical data.

Non-functional Requirements

The non-functional requirements outline the performance aspects essential for achieving our project goals:

- **Ease of Use:**

- The dashboard must feature an intuitive user interface and be accompanied by clear documentation for user guidance.

- **Performance:**

- The dashboard should exhibit responsive behavior with minimal loading times to ensure smooth user interactions.

- **Maintenance:**

- Code and data should be organized for easy maintenance, updates, and potential integration of new features.

- **Reliability:**

- The dashboard must consistently provide accurate and reliable stock analysis, ensuring uninterrupted functionality for users.

3.2.2 Global use case diagram

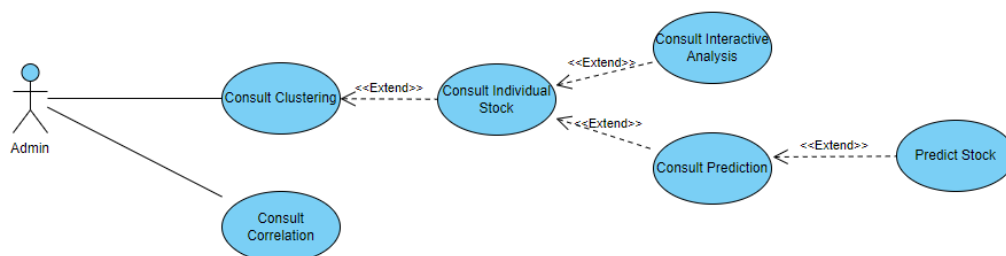


Figure 3.2: Global use case diagram

3.2.3 Refinement of the Use Cases

In this portion, we will display the refinement of each of our use cases.

Refinement of the use case Consult Clustering

Table 3.1, represents the refinement of the "Consult Clustering" use case.

Table 3.1: Description of use case "Consult Clustering"

Use Case	Consult Clustering
Actor	Admin
Precondition	Running system & available stock data
Nominal Scenario	1. The website shows the clustering page 2. The system displays the clustering data and chart and existant companies
Post-Condition	Clustering data is displayed successfully
Exception Scenario	The website shows an error message if there's no data

Refinement of the use case Consult Individual Stock

Table 3.2, represents the refinement of the "Consult Individual Stock" use case.

Table 3.2: Refinement of the use case "Consult Individual Stock"

Use Case	Consult Individual Stock
Actor	Admin
Precondition	Running system & available individual stock data
Nominal Scenario	1. The admin selects a specific company to look at its chart. 2. The system displays the individual stock data to the admin.
Post-Condition	Data is displayed successfully.
Exception Scenario	The system displays an error if the data is unavailable.

Refinement of the use case Consult Interactive Analysis

Table 3.3, represents the refinement of the "Consult Interactive Analysis" use case.

Table 3.3: Refinement of the use case "Consult Interactive Analysis"

Use Case	Consult Interactive Analysis
Actor	Admin
Precondition	Running system & available interactive analysis tools
Nominal Scenario	1. The admin navigates to the interactive analysis page. 2. The system retrieves data of multiple companies.
Post-Condition	Interactive analysis chart is displayed.
Exception Scenario	The system shows an error if the chart is not displayed.

Refinement of the use case Consult Ratios and performance

Table 3.4, represents the refinement of the "Consult Ratios and performance" use case.

Table 3.4: Refinement of the use case "Consult Prediction"

Use Case	Consult Ratios and performance
Actor	Admin
Precondition	Running system & available financial data
Nominal Scenario	1. The admin navigates to the consult prediction page. 2. The system retrieves data relevant to calculating the predictions.
Post-Condition	Stock Predictions are displayed on screen.
Exception Scenario	The system displays an error if the data is unavailable.

Refinement of the use case Predict Stock

Table 3.5, represents the refinement of the "Predict Stock" use case.

Table 3.5: Refinement of the use case "Predict Stock"

Use Case	Predict Stock
Actor	Admin
Precondition	Running system & historical stock data
Nominal Scenario	1. The admin selects the prediction option. 2. The admin inputs how many days to predict. 3. The system uses a trained model to show future days predictions.
Post-Condition	Predictions Chart is displayed.
Exception Scenario	The system displays an error if there was an issue with the financial data or the model.

Refinement of the use case Consult Correlation

Table 3.6, represents the refinement of the "Consult Correlation" use case.

Table 3.6: Refinement of the use case "Consult Correlation"

Use Case	Consult Correlation
Actor	Admin
Precondition	Running system & available financial data
Nominal Scenario	1. The admin navigates to the correlation analysis page. 2. the system retrieves relevant financial data and computes correlation coefficients.
Post-Condition	The correlation analysis results are displayed on screen.
Exception Scenario	The system shows an error if the data retrieved was not enough.

3.3 System Modeling

The following section presents the system modeling of the project.

3.3.1 Global class diagram

Here is the global class diagram which tells us about our data schema.

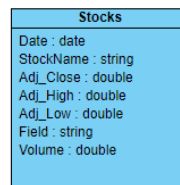


Figure 3.3: Global class diagram

In this class diagram , we have one class Stocks(Date , StockName , Adj-Close , Adj-High , Adj-Low , Field , Volume) that represents all the stock prices every day

3.3.2 Sequence diagrams of our application

Sequence Diagram of the use Case Consult Clustering

Figure 3.4, represents the sequence diagram of the "Consult Clustering" use case.

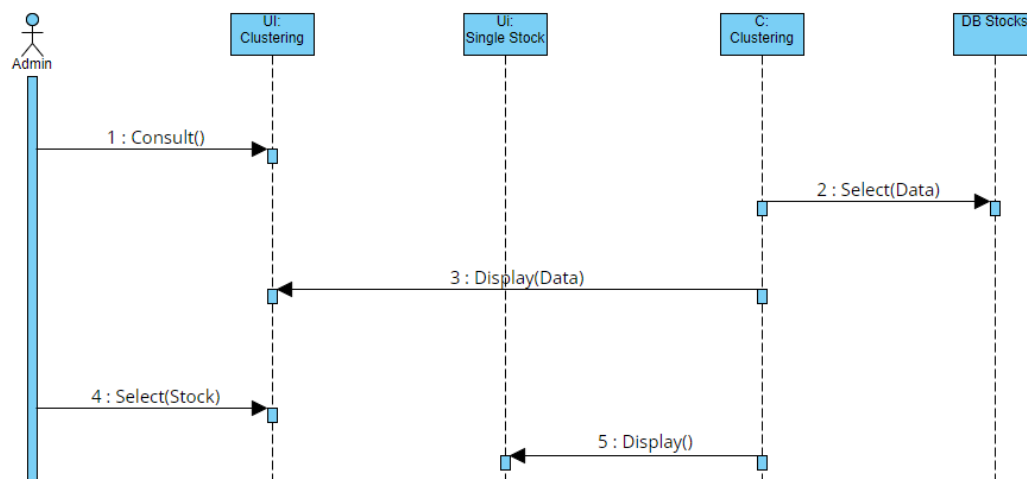


Figure 3.4: Clustering Sequence Diagram

Sequence Diagram of the use Case Consult Correlation

Figure 3.5, represents the sequence diagram of the "Consult Correlation" use case.

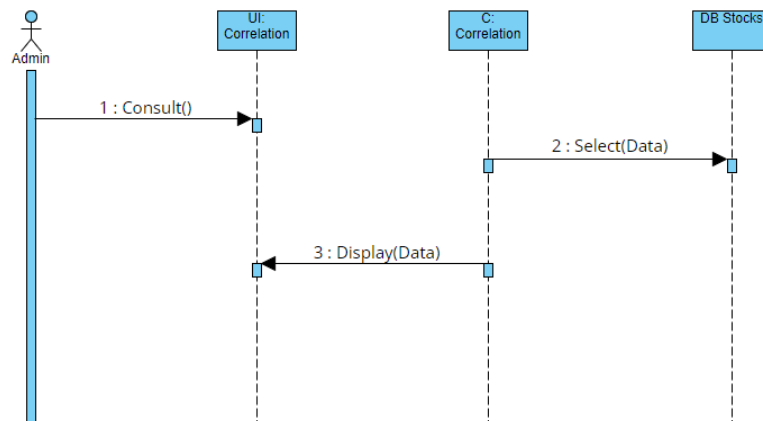


Figure 3.5: Correlation Sequence Diagram

Sequence Diagram of the use Case Consult Individual Stock

Figure 3.6, represents the sequence diagram of the "Consult Individual Stock" use case.

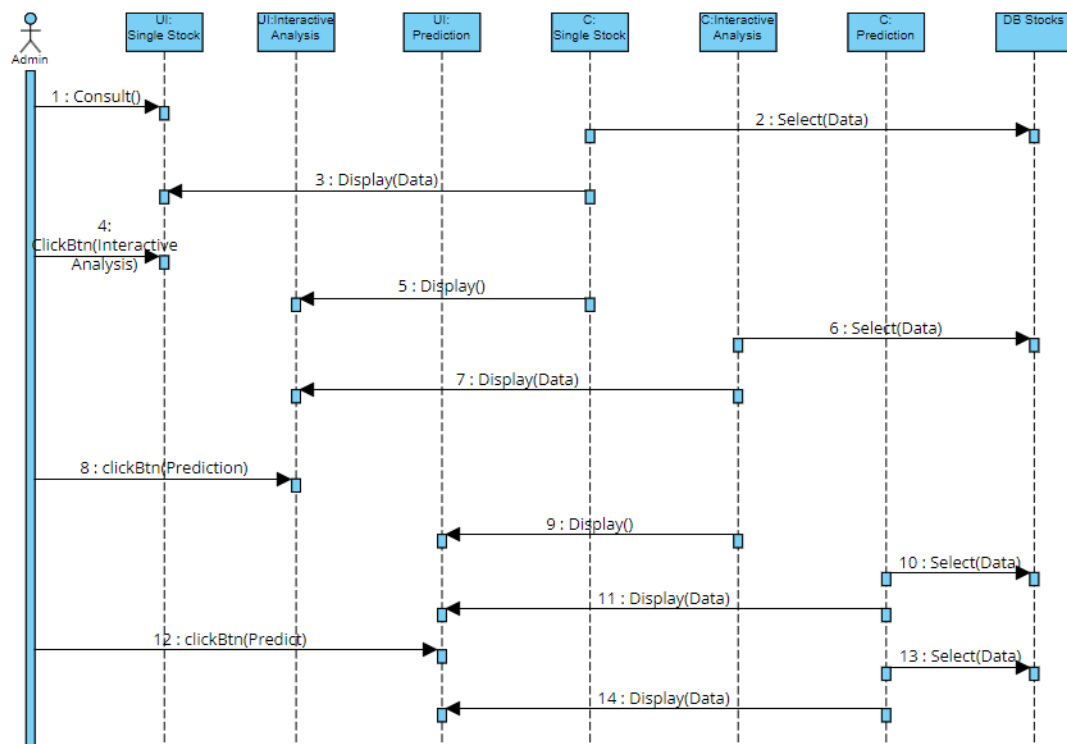


Figure 3.6: Individual Stock Sequence Diagram

3.4 Conclusion

This chapter has basically looked at how the system requirements and modelling in this case are important to stock forecasting web application. When you define specific functional as well as non-functional necessities you create a sturdy base for design that entails ease of use, efficiency, and continued relevance.

Chapter 4

Data Analysis and prediction

4.1 Introduction

Chapter 4 provides an in-depth exploration of stock price forecasting techniques. It begins by detailing the meticulous process of data collection, cleaning, and validation to ensure dataset reliability. Subsequently, various data analysis methods, including descriptive statistics, correlation analysis, and time series analysis, are employed to uncover underlying patterns in stock prices. The chapter then delves into the implementation of different forecasting models, such as linear regression, ARIMA, random forest, and LSTM, highlighting their methodologies and steps. Finally, a comparative analysis of these models' performance offers insights into their strengths and weaknesses in predicting stock prices.

4.2 Data Preparation

In the world of data science the most important step is data preparation which involves cleaning, transforming and organizing raw data in ways that are suitable for analysis or machine learning models. The following are some of the key steps in data preparation

4.2.1 Data Collection

Effective data preparation begins with thorough data collection. This section outlines the sources and methods used.

Source of Data

The data from the study was collected directly from a reputable financial institution that deals with stock market data analysis and trading.

Data Format

The host company structured the provided data in JSON format, a lightweight data interchange format that is popularly used to pass data between a server and a web application.

Data content

The host company provided JSON files that contained comprehensive data on stock market and had been subdivided into important elements such as:

- **Date:** The stock market data observations date is usually like YYYY-MM-DD and represent trading days
- **Stock Name:** The stock associated data observation name or ticker symbol. Eg; Apple Inc. as "AAPL"
- **Code_ISIN:** The stock related International Securities Identification Number (ISIN) code. The ISIN is a distinctive code for every security that allow international trading and settlement process.
- **Adj_Close:** The price at which the stock closed on a relevant date after various adjustments have been made. Adjustments could include stock splits, dividends or rights issues; adjusted closing price is therefore a better measure of what a share is really worth.
- **Adj_Low:** The low stock price on a specific date that represents the lowest trading price when adjusted for corporate actions.
- **Adj_High:** The high stock price on a specific date that represents the highest trading price when adjusted for corporate actions.
- **Volume:** The trading volume equals the total number of shares traded for the stock on the specific date. In other words, it represents how many shares have been exchanged between buyers and sellers throughout that period.
- **Field:** The field or sector in which the stock is categorized explains about the industry or market segment to which the stock belongs. This information provides context.

Data Parsing

After we found JSON text documents having stock trade particulars, we made use of Python's json library for parsing the documents in order for us to obtain important information. Therefore, we went through the structured data format of such documents with aim of capturing variables including stock date, stock symbol, price data like close prices ect... then processed each JSON object that represents an observed data retrieving the necessary fields it consists of into relevant structures to be applied later.

Conversion to DataFrame

We used Python pandas library to structure the extracted data after parsing and each data point which was in the form of a JSON object or as a dictionary got assimilated into DataFrame's row containing fields it was derived from (date, stock name, ISIN code, adjusted closing price, adjusted highest price, adjusted lowest price, trade volume amounts, and the field).

This change made it easy to handle, analyze and model data using DataFrame data format for flexible and functional purpose. The outcome was that DataFrame became a fundamental source of information for additional cleaning up, exploratory research, as well as modeling purposes in datascience process. In the figure 4.1 below is shown the data we are going to work with in this project.

	Date	Code_isin	Adj_Close	Stock_name	Adj_low	Adj_High	Volume
198870	2011-01-03		11.173	BNA	10.952	11.298	1721.199951
198814	2011-01-04		11.129	BNA	10.952	11.129	105979.023438
198758	2011-01-05		11.349	BNA	10.989	11.349	86125.226562
198702	2011-01-06		11.283	BNA	11.173	11.393	43161.000000
198646	2011-01-07		11.305	BNA	11.210	11.320	52695.449219
...	...		...	...	...	...	...
378	2021-10-29		7.340	BNA	7.300	7.340	6610.660156
299	2021-11-01		7.320	BNA	7.200	7.340	5781.819824
220	2021-11-02		7.310	BNA	7.200	7.320	24964.289062
141	2021-11-03		7.320	BNA	7.100	7.320	34433.000000
62	2021-11-04		7.320	BNA	7.220	7.320	6137.930176

Figure 4.1: Data Content

4.2.2 Data Cleaning

Data cleansing is a requisite procedure necessary to ascertain the quality and integrity of the dataset for purposes of analysis and modeling. Below, we look at the various data cleansing techniques that have been applied to this stock market data.

Handling Missing Values

There were several methods that were used to fix the missing values on the data set:

- **Forward Filling/Backward Filling:** Missing values in time-series data, such as stock prices, were filled using the preceding or succeeding non-missing value.

Feature Engineering

- **Lagged Variables:** Lagged versions of the target variable (stock prices) were created to capture temporal dependencies.
- **Rolling Statistics:** Rolling averages, standard deviations, and other statistical measures were computed to capture trends and patterns in the data.
- **Technical Indicators:** Technical indicators such as moving averages, Relative Strength Index (RSI), and MACD (Moving Average Convergence Divergence) were derived to incorporate market sentiment and momentum into the analysis.

Normalization and Scaling

Features were normalized or scaled using techniques such as Min-Max scaling or StandardScaler to ensure that they had similar scales and distributions.

Data Splitting

The dataset was split into training, validation, and testing sets to facilitate model training, validation, and evaluation, respectively. Care was taken to preserve the temporal order of the data to mimic real-world scenarios.

4.3 Data Analysis

By focusing on one stock only in this section, we are able to get deeper insights about how it performs and hence understand it better. Here we endeavour to narrow down our focus so as to identify meaningful trends and patterns which will help us make decisions based on information rather than luck. Deep analysis enables us pick useful information hence helping us understand what is happening in the stock market leading to development of efficient ways of investing.

4.3.1 Descriptive Statistics

Descriptive statistics summarize the typical values, dispersion, and form of a collection of information. Use tools employed to click out such numbers includes the average, median, deviation from mean, smallest, and largest scores. In the figure 4.2 is represented all these values.

Descriptive Statistics:					
	Date	Adj_Close	Adj_low	Adj_High	\
count	2689	2689.000000	2689.000000	2689.000000	
mean	2016-06-10 02:32:05.176645632	8.846972	8.517569	8.678691	
min	2011-01-03 00:00:00	5.053000	0.000000	0.000000	
25%	2013-09-27 00:00:00	7.115000	7.017000	7.121000	
50%	2016-06-08 00:00:00	8.223000	8.076000	8.251000	
75%	2019-02-19 00:00:00	9.916000	9.818000	9.999000	
max	2021-11-04 00:00:00	19.799000	19.356000	19.889000	
std	NaN	2.457952	2.853659	2.915732	
Volume					
count	2.689000e+03				
mean	1.708736e+05				
min	0.000000e+00				
25%	1.208680e+04				
50%	4.341140e+04				
75%	1.322846e+05				
max	1.801664e+07				
std	6.216704e+05				

Figure 4.2: Descriptive Statistics of BNA stocks

Benefits: This provides a summary of the important statistical measures hence assisting in perceiving the mean, spread and pattern of the data.

Prediction Benefit: This assists in recognizing any patterns that could be emerging or help in spotting any potential trends from previous prices movements which had occurred before.

Interpretation: Descriptive statistics helped us understand the stock's central tendency and volatility. This analysis guided our choice of models, such as LSTM, to handle varying data stability and risk.

4.3.2 Correlation Analysis

Correlation analysis is used to investigate the relationship between two or more variables contained in a given dataset by determining correlation coefficients which reveal the magnitude and direction of linear relationships. In the figure 4.3 below we have the Correlation Matrix of our data.

Correlation Matrix:				
	Adj_Close	Adj_low	Adj_High	Volume
Adj_Close	1.000000	0.897132	0.898210	0.253501
Adj_low	0.897132	1.000000	0.999222	0.230932
Adj_High	0.898210	0.999222	1.000000	0.237956
Volume	0.253501	0.230932	0.237956	1.000000

Figure 4.3: correlation of BNA Stocks

Benefits: Helps one see the power and direction of relationships connecting variables and identify likely predictors of stock price changes.

Prediction Benefit: This understand how various financial measures correlate with stock prices enables analysts to give priority to those factors that are closely linked when carrying out predictive modeling .

Interpretation: Correlation analysis revealed relationships between financial variables. It informed our feature selection, ensuring the inclusion of relevant variables in models like multivariate linear regression and Random Forest.

4.3.3 Time Series Analysis

Time series analysis is the process of examining patterns, trends, and seasonality within a dataset over time. It requires looking at consecutive data points that have been gathered at fixed intervals. Below is the figure 4.4 representing the time series analysis of the BNA adjusted close price

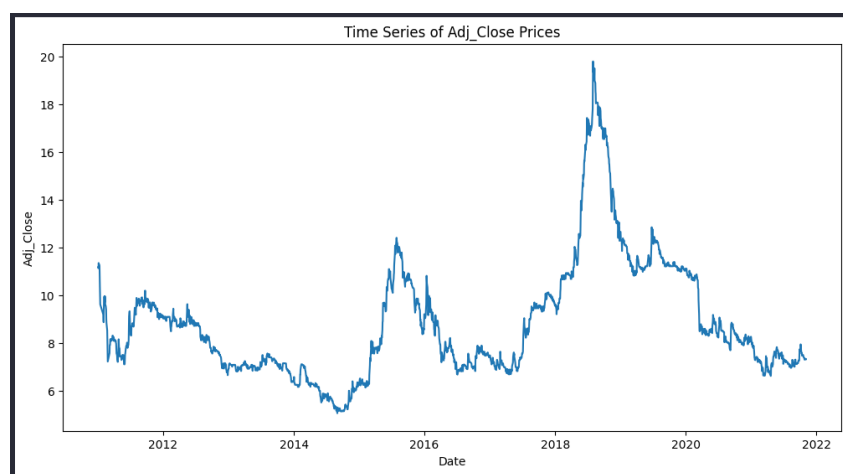


Figure 4.4: Time series Analysis of Bna Adjusted close price

Benefits: It allows patterns, trends, and seasonality identification in the dataset to help us get an idea on how stock price movement happens over time.

Prediction Benefit: The prediction benefit is that historic events can be extrapolated for the purpose of predicting future price movements in stocks.

Interpretation: Time series analysis identified patterns and trends over time, leading to the choice of ARIMA and LSTM models, which handle time-dependent data effectively.

4.3.4 Distribution Analysis

Analysis on distribution ventures into where within a dataset data values are distributed and uses histograms or density-plot to visualize frequency distribution of numerical variables. Below the figure 4.5 and the figure 4.6 represents the distribution analysis of the BNA adjusted price and volume

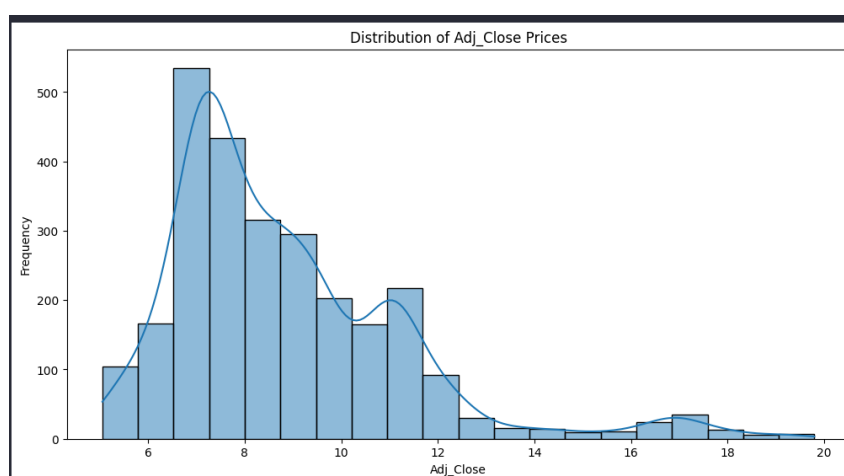


Figure 4.5: Distribution Analysis of BNA Adjusted close price

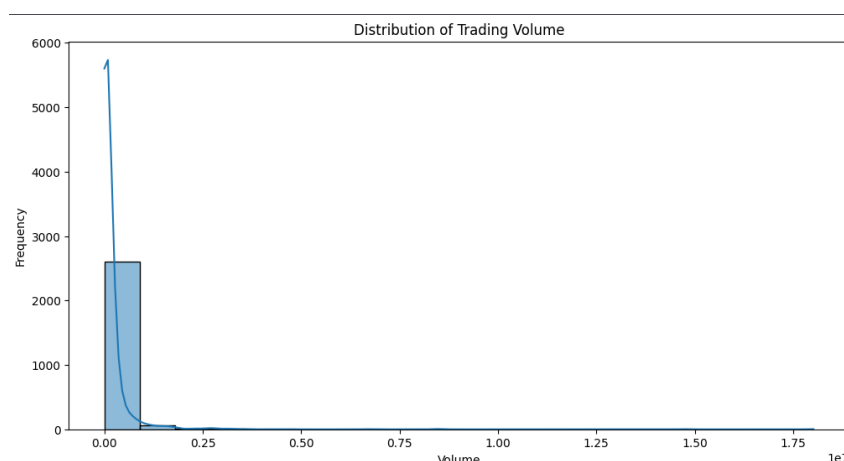


Figure 4.6: Analysis of BNA volume

Benefits: When used, it displays the frequency distribution of numeric variables; therefore, makes it easier to know how far apart these particular data points are from each other.

Prediction Benefit: This assists in determining how large/small possible changes in share prices could be at some time point in future. This also helps to determine certain assumptions concerning stock price risks.

Interpretation: Distribution analysis provided insights into the frequency distribution of stock prices. It helped normalize the data and guided the selection of models like Random Forest and LSTM that handle non-normal distributions well.

4.3.5 Outlier Detection

Outlier detection is responsible for noticing critical observations that vary significantly from other data. Consequently, it assists in noticing anomalies or errors in the dataset that solicit further investigation. In the figures 4.7 and 4.8 we plotted the outlier detection on BNA close prices and volume .

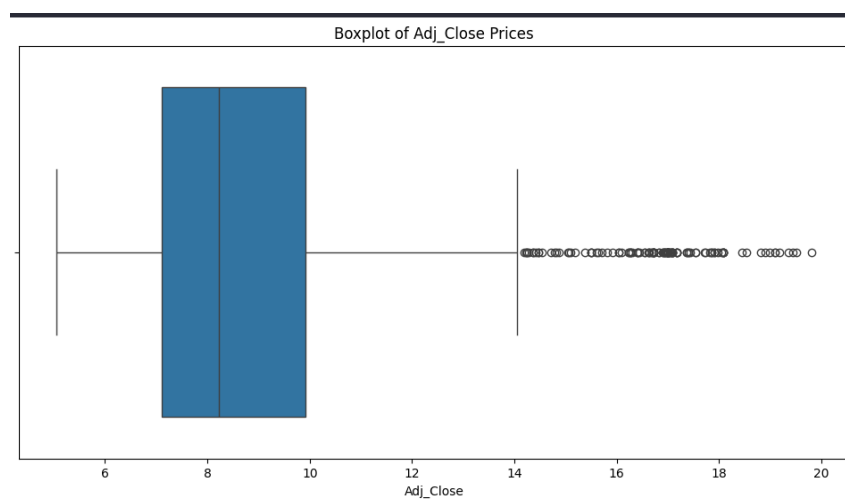


Figure 4.7: Outlier detection on BNA close prices

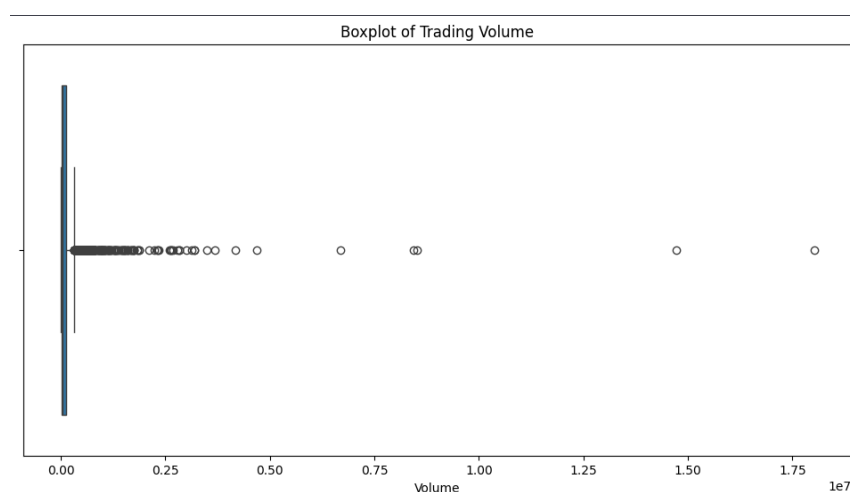


Figure 4.8: Outlier detection on BNA volume

Benefits: This detects any strange data points that are far from being typical and possibly wrongful or strange values or very rare events.

Prediction Benefit: Outliers that distort predictions can be removed and thus clean up the dataset so as to make it more accurate in predictive modeling.

Interpretation :Outlier detection ensured a cleaner dataset by identifying and addressing anomalies. This step improved the reliability of models, particularly linear regression, which are sensitive to outliers.

4.3.6 Moving Average Analysis

The moving average analysis determines the average value of a quantity within a certain period. It is useful in ensuring regular amounts against changes and revealing invisible dynamics in the data. Below the figure 4.9 representing the moving average of the BNA stock.

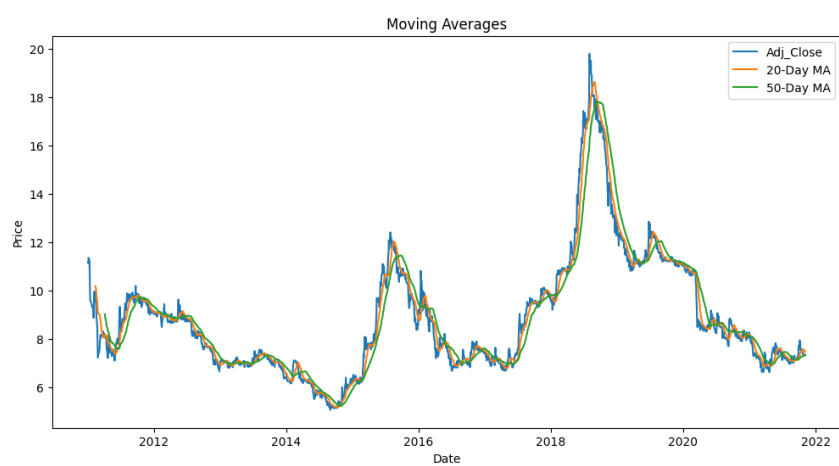


Figure 4.9: Moving Average BNA

Benefits: Reverses unstable conditions that are influenced by shifts that take place in the short term, thus bringing to light the actual patterns and directions evident in a particular set of information.

Prediction Benefit: Clearly defines the general trend which can be speculated to tell what will be the future price changes.

Interpretation :Moving average analysis highlighted long-term trends in stock prices, informing the use of models like ARIMA and LSTM that can capture and forecast these trends.

4.3.7 Volatility Analysis

Volatility analysis determines how much a data set changes or varies by time. It shows how much the prices on financial markets change a lot, thus giving a hint of the risks involved. Below is shown the volatility of our bna data (figure 4.10).

Annualized Volatility: 0.23331851423007274

Figure 4.10: BNA volatility

Benefits: It helps to measure variance or unpredictability in stock prices which is useful when assessing risks and managing portfolios in such ranges where different assets classes are available .

Prediction Benefit: It mainly assists in expecting how much prices might change tomorrow by considering how unstable current markets are and then deciding on appropriate trading approaches.

Interpretation :Volatility analysis measured price fluctuations, guiding the choice of robust models like Random Forest and LSTM to manage high variability and provide stable predictions.

4.3.8 Seasonality Analysis

Seasonality analysis help you identify patterns that repeat in datasets over given intervals of time like daily, weekly or monthly periods.In the figure 4.11 below is the plots of the seasonality analysis.

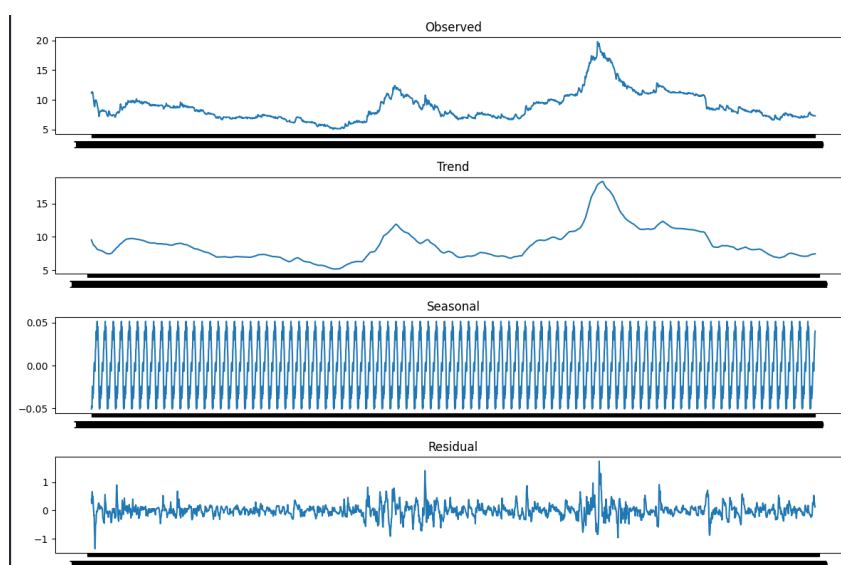


Figure 4.11: Seasonality Analysis of the BNA

Benefits: By making use of seasonal effects and calendar components, it is easy to recognize repetitive patterns and cycles present in the data .

Prediction Benefit: Application in forecasting may be examined in terms of seasonal trends, making stock price forecast more accurate in the subsequent periods.

Interpretation :Seasonality analysis identified recurring patterns in stock prices, leading to the use of ARIMA and LSTM models that incorporate seasonal trends for more accurate predictions.

4.3.9 Relative Strength Index (RSI)

It's very important for traders to know the Relative Strength Index (RSI) since it tell them how past trends of stocks have performed in terms of momentum. More specifically, this technical indicator shows whether an asset has reached its peak yet it's still rising up some minutes later (overbought) or it has already reached rock bottom yet something

keeps on pulling it back (oversold) so that buy or sell recommendations can be given precisely at those times. Below is the figure(4.12) representing the RSI of the BNA stock.

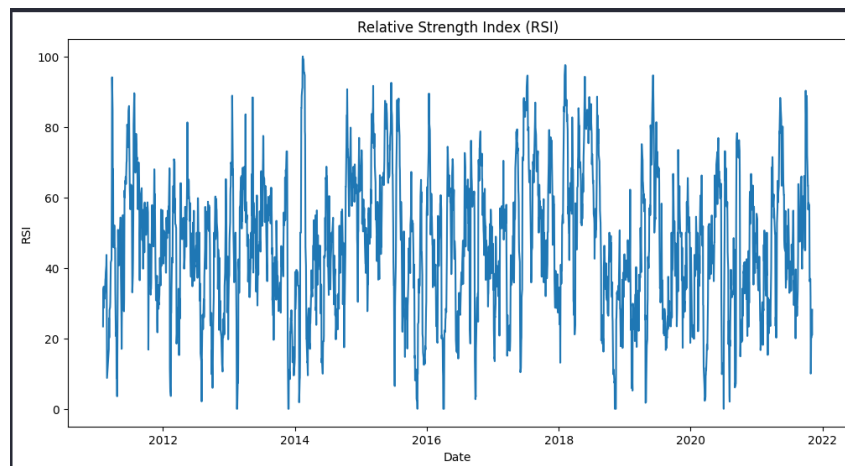


Figure 4.12: BNA stock RSI

Benefits: Assessing the recent amount and speed at which prices have changed. This in turn provides an indication on whether a stock is overbought or oversold.

Prediction Benefit: Predictive capability that gives directional clues for possible impending trend reversals, hence allowing investment advisors anticipate changes that are likely to occur to a company's share value depending on momentum indicators

Interpretation :RSI analysis indicated whether a stock was overbought or oversold, improving timing for predictions. Incorporating RSI into models, especially Random Forest, enhanced their predictive capabilities.

4.3.10 Price-Volume Analysis

Price-volume analysis is the examination of how stock price moves in relation to its trading volume. It is used to determine the trend's power in pricing and to recognize any turn-around in the market. Below the representation of the close price and the volume (figure 4.13).

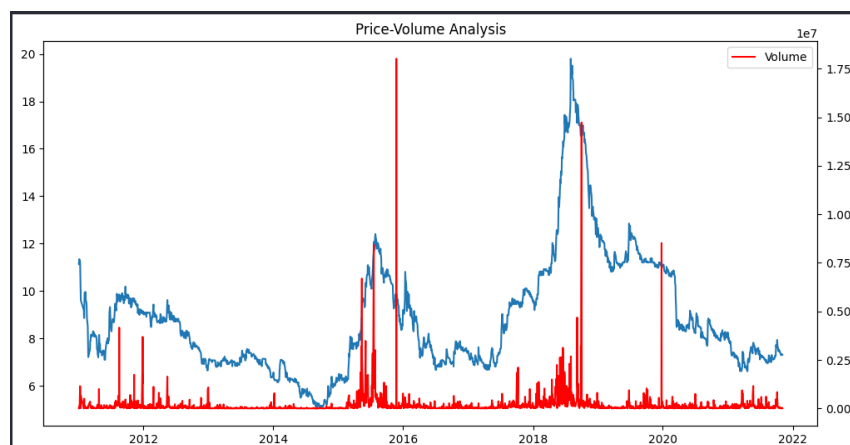


Figure 4.13: Price-Volume Analysis of BNA

Benefits: One aspect investigates stock prices in correlation with trade volumes which can shed light on trends' power or possibility for continuation while posterity.

Prediction Benefit: it can help us know whether the movements on prices are really valid ones; or put in other words, its aim is to signal potential market reversals by observing changes occurring in trading volumes about any asset's price/frequency ratio.

Interpretation :Price-volume analysis helped understand the strength of price movements and potential market reversals. It enriched our feature set, improving the performance of models like Random Forest and LSTM.

4.3.11 Interpretations and Conclusions

These analyses informed our decisions on model implementation and data preprocessing for optimal performance. Each analysis provided insights that shaped our approach:

- **Descriptive Statistics & Distribution Analysis:** Guided data normalization and feature scaling.
- **Correlation & Time Series Analysis:** Informed feature selection and favored time-dependent models.
- **Outlier Detection:** Ensured data integrity, leading to more accurate model training.
- **Volatility & Seasonality Analysis:** Helped select models like ARIMA and LSTM that handle high variability and periodic trends.
- **RSI & Price-Volume Analysis:** Added features that enhanced machine learning model predictions.

Overall, these analyses were crucial in understanding stock market data and selecting the best models for accurate stock price prediction.

4.4 Model Implementation

This section details the implementation of various predictive models for stock analysis.

4.4.1 Simple Linear Regression

Simple Linear regression offers a straightforward approach to predicting stock prices based on historical data.

Model Implementation

An included feature in this release was additional engineering on the characteristics which increased the predictability of the model.

Data Preparation:

- Cutting data into training and testing sets involved using the `train_test_split` function.

- Features (X) was made for the stock market information reshaped into the index value. (`Adj_Close`) as the target variable (Y) .

Model Training:

- We standardized the feature data with `StandardScaler`.
- We imported and established the `LinearRegression` model.
- The `fit` method was utilized to train the model on training data.

Below (figure 4.14) is the plot of the actual vs the predicted values of the simple linear regression.

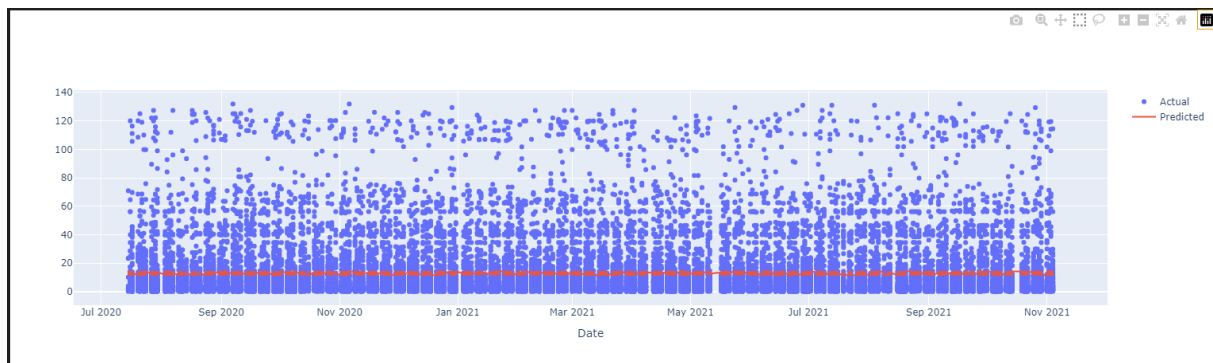


Figure 4.14: Plot actual and predicted values Linear Regression Version 1

Model Evaluation:

For model evaluation, score was computed using the subsequent metrics:

Metric	Train	Test
R^2	[<code>r2_score_train</code>]	[<code>r2_score_test</code>]
MSE	[<code>mse_train</code>]	[<code>mse_test</code>]

and we got those values shown in the figure below (figure 4.15) :

Metric	Train	Test
<code>r2_score</code>	0.0025035055829816244	0.0023651456090634237
<code>MSE</code>	402.8027345469547	405.5839827625187

Figure 4.15: Linear Regression v1 evaluation

Based on our experiments, we realized that in volatile markets, such as one experienced in the stock prices as at current time, it is not possible to adequately predict future price movements with a simple linear regression model.

4.4.2 Multivariate Linear Regression

Just like the simple linear regression it uses historical data but uses more features

Model Implementation

In this edition, some extra characteristics were considered to improve the forecasting abilities enshrined on the straight line regression model.

Feature Engineering: In this version, we engaged in extensive feature engineering for enhancing the dataset with informative variables that could potentially enhance predictive performance of the linear regression model.

- **Daily Returns:** Stocks' daily returns indicate the percentage by which stock prices change from a day's trade to that of another. The volatility and directionality of stock prices' moves are what daily returns reveal in the long run.
- **Simple Moving Averages (SMA):** We have calculated simple moving averages (SMA) for different windows lengths including 10 days, 20 days and 50 days. These averages are calculated by summing the closing prices over that time frame and then dividing by the number of periods to obtain average price which makes underlying patterns more visible in stock market fluctuations.
- **Volume Shocks:** We introduced a binary variable to indicate volume shocks in the stock market data. A volume shock occurs when the trading volume significantly deviates from its previous value, signaling potential changes in market sentiment or liquidity. We represented volume shocks as 1 for an increase in volume and 0 otherwise.

Data Preprocessing:

- The training and testing sets were obtained by splitting the dataset with the `train_test_split` function.
- With `SimpleImputer`, we applied a mean strategy to missing values within the training set..
- `StandardScaler` was used to normalize feature data..

Model Training:

- We imported and instantiated the `LinearRegression` `LinearRegression` model.
- The training data was used to train the model through the `methodfit`.

Here in this plot (figure 4.16) we compared the actual close prices values vs the predicted close prices of our model

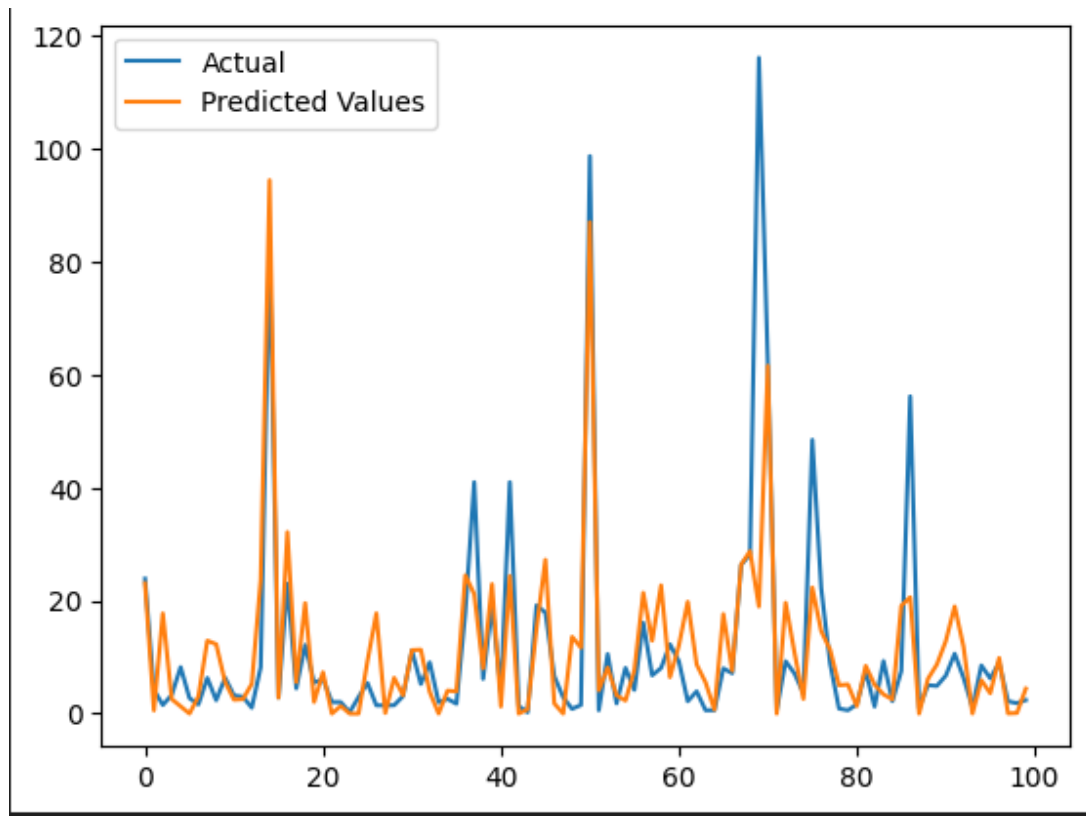


Figure 4.16: Actual vs predicted close prices plot of Multivariate Linear regression

Model Evaluation:

We calculated how accurate is this model by using the following evaluation metrics; mean square error(MSE), mean average error(MAE) and in question raising square root(MSE). And we got these values :

- Mean absolute Error : 6.6588171924655875
- Mean squared Error : 140.3586508323016
- Root Mean Squared Error : 2.5804684056321223

We found that the Multivariate Linear Regression is ineffective as it assumes all the variables to be linearly related , leading us to wrong predictions especially in scenarios which involves complex or noisy data sets.

4.4.3 ARIMA Model Implementation

Arima models are powerful tools for time series forecasting and to capture temporal dependencies.

Data Preparation

We imported JSON stock market data into a pandas Dataframe, extracted data about BNA stock and ordered it by date. We also indexed prices by date to make sure time series analysis easily becomes possible.

Model Building

We built the ARIMA model using the ARIMA class named `statsmodels` library. The ARIMA model's order was determined by its autocorrelation function (ACF) and partial autocorrelation function (PACF) plots as (7,0,6). In the figure 4.17 below is the results of our Arima model

SARIMAX Results						
=====						
Dep. Variable:	Adj_Close	No. Observations:	2689			
Model:	ARIMA(7, 0, 6)	Log Likelihood	1597.897			
Date:	Mon, 29 Apr 2024	AIC	-3165.794			
Time:	22:18:41	BIC	-3077.340			
Sample:	0	HQIC	-3133.800			
	- 2689					
Covariance Type:	opg					
=====						
	coef	std err	z	P> z	[0.025	0.975]

const	8.9045	0.921	9.672	0.000	7.100	10.709
ar.L1	0.9115	0.536	1.700	0.089	-0.139	1.962
ar.L2	0.4613	0.354	1.305	0.192	-0.232	1.154
ar.L3	0.1590	0.453	0.351	0.726	-0.729	1.047
ar.L4	-0.3257	0.345	-0.945	0.345	-1.001	0.350
ar.L5	-0.2908	0.269	-1.083	0.279	-0.817	0.236
ar.L6	0.1014	0.252	0.402	0.687	-0.392	0.595
ar.L7	-0.0170	0.212	-0.080	0.936	-0.433	0.399
ma.L1	0.2471	0.537	0.460	0.645	-0.805	1.299
ma.L2	-0.3220	0.410	-0.786	0.432	-1.125	0.481
ma.L3	-0.6193	0.169	-3.670	0.000	-0.950	-0.289
ma.L4	-0.3240	0.326	-0.995	0.320	-0.962	0.314
ma.L5	0.0504	0.332	0.152	0.879	-0.600	0.701
...						
=====						

Figure 4.17: Arima model build summary

Model Evaluation

To assess the coefficients and statistical significance of the ARIMA model, we have printed the model summary. On top of that, tomorrow's stock price was predicted by means of the forecast method, including confidence intervals (Forecast method). In the figure 4.18 we have the plot of the actual vs predicted close prices of this model

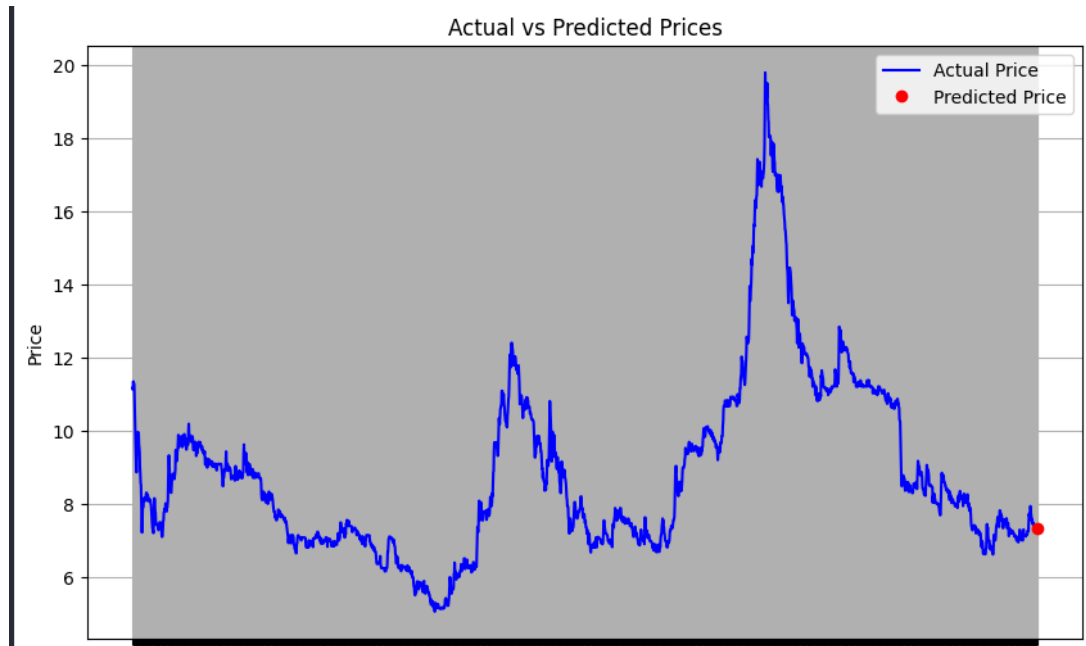


Figure 4.18: Arima pLot of predicted day

Performance Evaluation

To evaluate the performance of the ARIMA model, the mean squared error (MSE), mean absolute error (MAE), root mean squared error (RMSE), and the coefficient of determination were used. The predicted and observed prices were shown graphically for the model's accuracy.

Here are the values :

Mean Squared Error (MSE): 3.670659429023053e-05

Mean Absolute Error (MAE): 0.006058596726159493

Root Mean Squared Error (RMSE): 0.006058596726159493

Based on our experiments we can say that the ARIMA model has good predictive qualities but it does not always hold true in highly volatile markets since the assumption of stationarity and linearity are violated leading to inaccurate forecasts and lack of detection about time trends.

4.4.4 Random Forest Model Implementation

Here we tried the Random Forest model known for its high accuracy .

Data Preparation

Just like the ARIMA model, the stock market data was brought into and saved in a pandas DataFrame. Feature columns (Adj_low, Adj_High, Volume) were chosen as the independent variables while the target variable (Adj_Close) was selected as the dependent variable for regression.

Model Building

We instantiated and trained a model for a RandomForestRegressor using chosen features as well as target values. Having trained our model, it was used for forecasting stock prices on the next business day according to aforementioned features.

Model Evaluation

The model's performance was evaluated using various evaluation metrics like MSE, MAE, RMSE, and R^2 . These evaluation metrics were computed by making use of the predicted and actual values of the next day's stock price. The Evaluation Metrics are :

Mean Squared Error (MSE): 0.0004161599999999808

Mean Absolute Error (MAE): 0.020399999999999953

Root Mean Squared Error (RMSE): 0.020399999999999953

R-squared (Coefficient of Determination): Can't be calculated

Visualization

We plotted the predicted prices against the date to visualize how the model performed over time. For further evaluation of model accuracy, an actual vs. predicted scatter plot was produced. Here is the scatter plot in the figure 4.19:

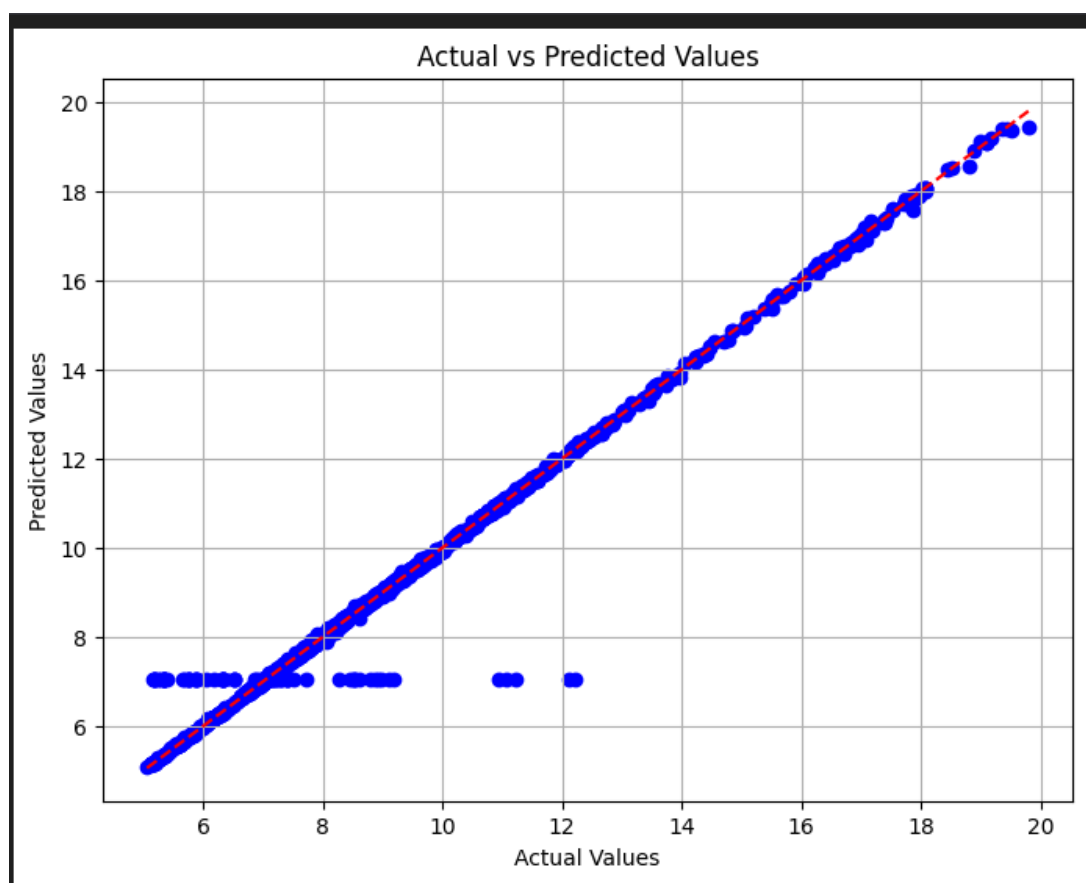


Figure 4.19: Random Forest Scatter plot close prices and actual prices

And here is the prediction of one future day in the figure 4.20:

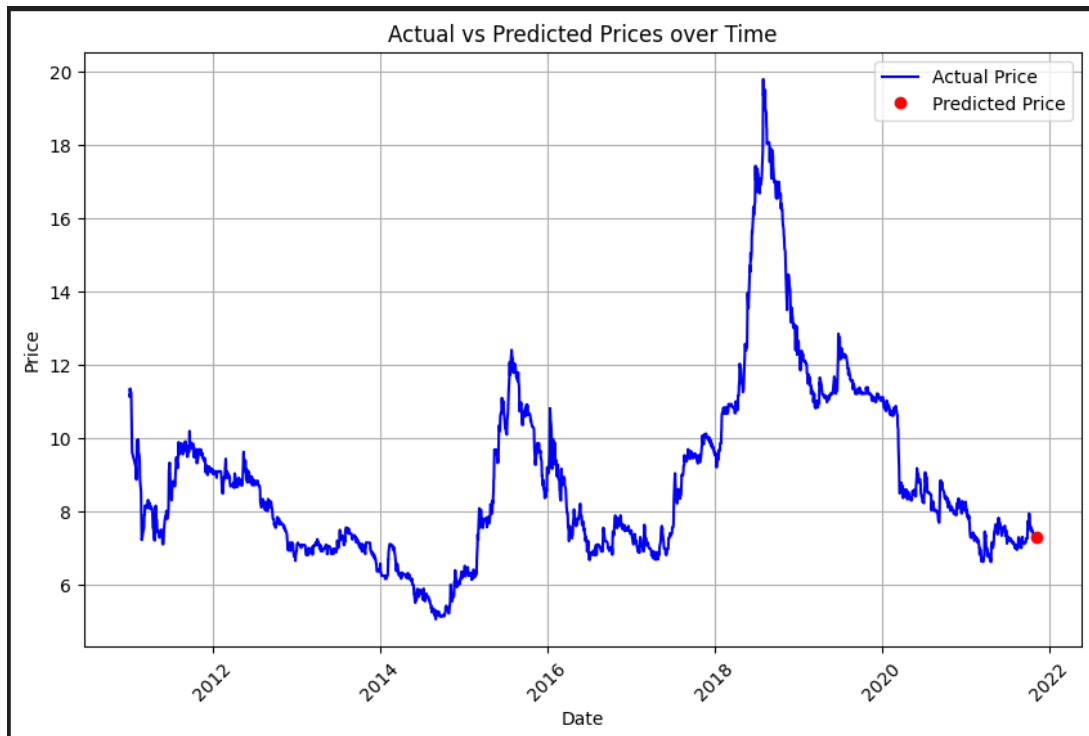


Figure 4.20: Random Forest future day prediction plot

Based on our experimentation the Random Forest Model, though an ensemble method, faced problems in detecting intricate patterns and preventing overfitting of noisy data.

4.4.5 LSTM Model Implementation (First Version)

We implemented the first version of our Long Short Term Memory Model

Data Preparation

The historical price of different stocks was contained in a JSON document which enabled the accurate retrieval and illustration of the stock markets files. Futile columns for example “Code isin” and “stock name” were also removed while datetime format has been transformed from date feature so has been marked as time.

Windowed Data Creation

In order to train the LSTM model, the data had to be prepared. Thus, a windowing technique was used. In this case, it involved shifting the prices of ‘Adj Close’ by certain number of days (for example, 3 days) in order to create input-output pairs.

Thus an input window would always consist of some past ‘n’ days’ prices while the output was always equal to the price on the next day, which then produced sequences of these input-output pairs that were used to train the model.

Here is the table of the windowed data where the **target** column is the price to predict(output) and the **target-1** and **target-2** and **target-3** are the past 3 days and act as the input to our model. As shown in this figure (4.21).

	Target-3	Target-2	Target-1	Target
Date				
2011-01-03	6.250	6.263	6.257	6.257
2011-01-04	6.388	6.250	6.263	6.257
2011-01-05	6.145	6.388	6.250	6.263
2011-01-06	5.961	6.145	6.388	6.250
2011-01-07	5.790	5.961	6.145	6.388
...	...	...	...	...
2021-10-26	11.900	11.900	11.900	11.900
2021-10-27	12.200	11.900	11.900	11.900
2021-10-28	12.200	12.200	11.900	11.900
2021-10-29	12.200	12.200	12.200	11.900
2021-11-01	12.200	12.200	12.200	12.200
2686 rows × 4 columns				

Figure 4.21: Windowed Data

Model Architecture

The model has an input layer that receives 3D tensor data. Afterwards it contains an LSTM layer of 64 units which is responsible for sequential data processing and maintains dependencies over time using hyperbolic tangent activation function. Meanwhile, two dense layers come next having 32 neurons each and activated by Rectified Linear Unit (ReLU) to introduce non-linearity that can help to identify complex patterns while avoiding overfitting by providing some sort of noise or occlusion during training phase.

Finally there is an output layer with only one neuron but having linear activation so as to produce the final predictions required. To optimize Mean Squared Error loss, training uses Adam optimizer with 0.001 of a learning rate, meanwhile Mean Absolute Error is used to evaluate it. The LSTM's ability to capture temporal patterns and dense layers that extract characteristics are utilized to tailor this architecture tailored for times series analysis.

Model Evaluation

Upon training the long short-term memorization model (LSTM) was tested with various datasets - training, validation as well as testing data - to gauge its performance across different dimensions. In order to evaluate how precisely the model works some performance metrics that included MSE and MAE were calculated. Furthermore, its predictions were checked against the real prices while the differences were illustrated through graphs showing its effectiveness at different times. Here are the evaluations metrics :

- Mean Squared Error (MSE): 0.40216389298439026

- Mean Absolute Error (MAE): 0.5608488321304321
- Root Mean Squared Error (RMSE): 0.6341639161109924
- R-squared (Coefficient of Determination): 0.730399665893748

Here are some plots to visualize and evaluate the model :

Training Predictions vs Training Observation

This plot (figure 4.22) shows the training prediction vs The actual training values



Figure 4.22: Training Predictions vs Training Observation

Validation Predictions vs Validation Observations

This plot (figure 4.23) shows the Validation prediction vs The actual Validation values

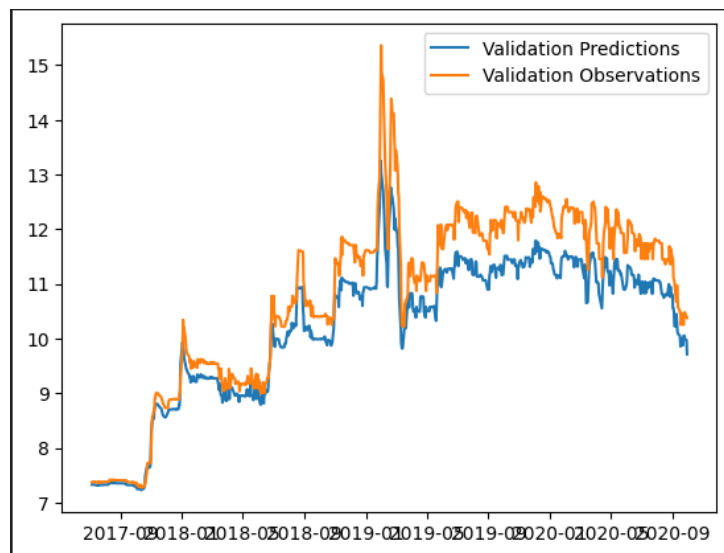


Figure 4.23: Validation Predictions vs Validation Observations

Test Predictions vs Test Observations

This plot (figure 4.24) shows the Test prediction vs The actual Test values

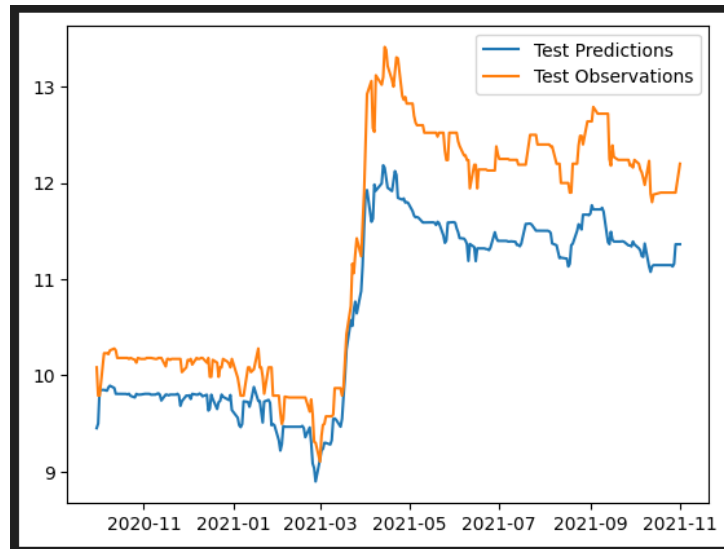


Figure 4.24: Test Predictions vs Test Observations

All of the plots above in the same plot

This plot (figure 4.25) concats all the plots shown before into one plot

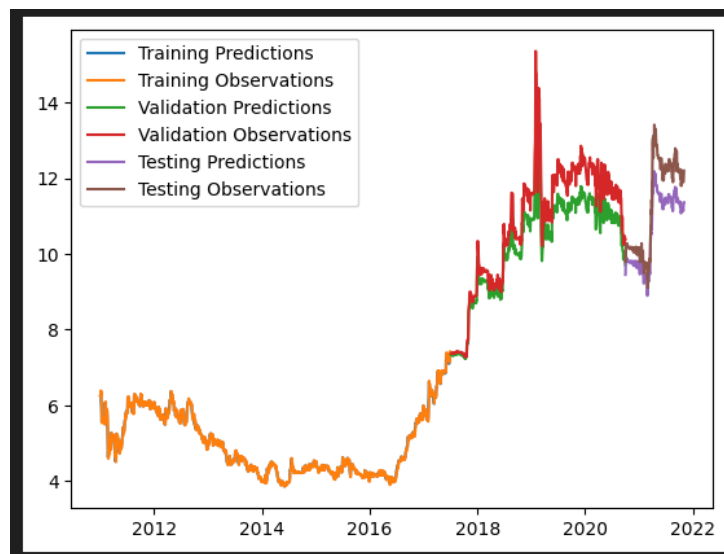


Figure 4.25: Prediction of close prices vs Observations

The previous plot with the future prediction

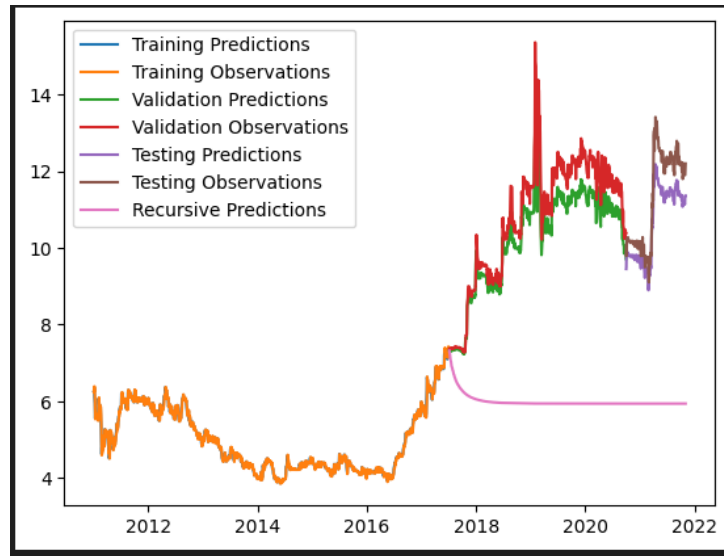


Figure 4.26: Prediction of close prices vs Observations & future close prices

In our experiments, the LSTM model First Version showed evidence of overfitting to the training data, with long-term dependencies not being well captured making it necessary to pay close attention when tuning hyperparameters to avoid this and for getting accurate forecasts.

4.4.6 LSTM Model Implementation (Second Version)

Here is our last model implementation , the Second Version of the Long Short Term Memory model

Data Preparation

In the same way as the initial one, the data preparation steps comprised importing the stock market data for the 'BNA' stock, filtering relevant columns, and setting the index as the 'Date' column. However, this version included extra pre-processing steps like using MinMaxScaler for normalization in order to scale data between 0-1 range. Such pre-processing refers to making sure that the model converges and performs better while training LSTM.

Model Architecture

The model architecture is like this: four levels of long-short term memory with dropputs following each layer. The layers are arranged in increasing order of secreted products from 50 through 120; they are all driven by rectified linear units (RLUs). In regard to temporal preservation, the first three layers pass sequences on whereas the last one followed by nondense units makes predictions. The model has been compiled with an optimizer called Adam and it computes Mean Squared Error loss. The test data is used for evaluation at the end of 50 epochs of training.

To assess model performance, Mean Squared Error (MSE), Mean Absolute Error (MAE),

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 100, 50)	10,400
dropout (Dropout)	(None, 100, 50)	0
lstm_1 (LSTM)	(None, 100, 60)	26,640
dropout_1 (Dropout)	(None, 100, 60)	0
lstm_2 (LSTM)	(None, 100, 80)	45,120
dropout_2 (Dropout)	(None, 100, 80)	0
lstm_3 (LSTM)	(None, 120)	96,480
dropout_3 (Dropout)	(None, 120)	0
dense (Dense)	(None, 1)	121

Figure 4.27: Summary architecture LSTM v2

Root Mean Squared Error (RMSE), and R-squared are computed lastly. Prediction accuracy is checked by visualizing actual along with predicted prices.

Here is the summary of the architecture in the figure 4.27;

Model Training and Testing

The Adam optimizer and mean squared error loss function were used to train a second version of the LSTM model over preprocessed data. Training lasted for 50 epochs with 32 as the batch size. The performance of the model was then evaluated on another separate dataset to determine whether it can generalize well. Such performance measures as MSE, MAE, and RMSE were calculated in order to estimate this model's accuracy against the first version.

Performance Evaluation

After going through training and testing, evaluations concerning the LSTM model were made with respect to several performance indicators; Mean squared error, Mean absolute error, Root mean squared error, and R-squared which is also known as Coefficient of Determination. They pretty much tell us how accurate our predictions are and whether or not they fit underlying structures of data that we can rely on. Additionally, these prices were visually matched against predictions made by the model via graphs hence enabling qualitative judgment of the model's performance in time.

Here are the evaluations metrics :

- Mean Squared Error (MSE): 0.021626693513499027
- Mean Absolute Error (MAE): 0.10121010684967029
- Root Mean Squared Error (RMSE): 0.14706016970444113
- R-squared (Coefficient of Determination): 0.5150779395956218

Here is the 100 day prediction vs 100 day real close price in the figure 4.28:

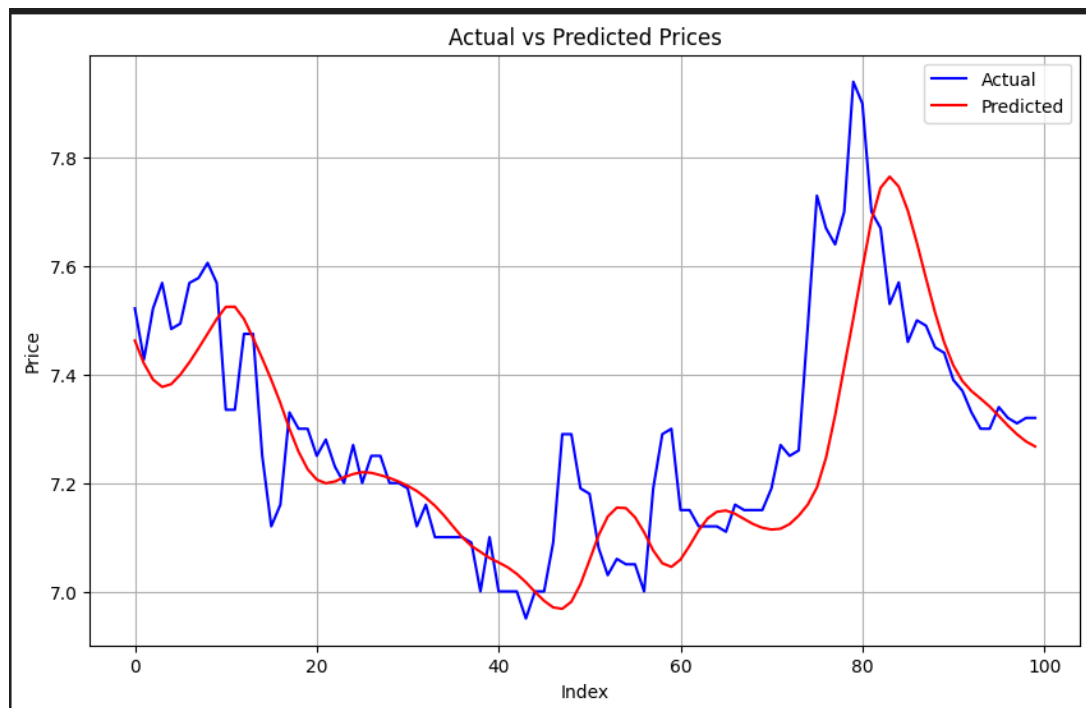


Figure 4.28: 100 day prediction vs 100 day real close price

This Model is so far the one that gave us the best result , it still struggles with minute detail in capturing stock value trends but it got very promising results.

4.5 Model Comparison

This section scrutinizes the performance and attributes of diverse models that have been deployed in forecasting stock prices. In particular, those that have been taken into account include Linear Regression (First Version), Linear Regression (Second Version), ARIMA, Random Forest, LSTM (First Version), as well as LSTM (Second Version) among others. Each one has its own special characteristics which makes it optimized both to capture certain peculiarities of stock market dynamics and determine subsequent changes in prices.

4.5.1 Comparison Table

Here's a comparative summary of the performance metrics obtained for each model in the table 4.1 below:

Table 4.1: Performance metrics for different models

Model	MSE	MAE	RMSE	R^2
Simple Linear Regression	402	-	-	0.0025
Multivariate Linear Regression	140.35	6.66	2.58	-
ARIMA	3.67e-05	0.00606	0.00605	-
Random Forest	0.000416	0.02	0.02	-
LSTM (First Version)	0.402	0.56	0.6	0.73
LSTM (Second Version)	0.02	0.1	0.14	0.51

Below in the table 4.2, we will explain the characteristics of each model by discussing what the model's method is, what pre-processing have been done, how this model was trained and which evaluation metrics have been used for this purpose.

Table 4.2: Models and Descriptions

Model	Description
Simple Linear Regression	Uses linear regression analyses for predictions. This may require some scaling of the features or other preliminary processing. Develops a linear regression model with the training set. Often, the assessment is made through metrics such as Mean Squared Error (MSE) or R-squared. No predictions are actually being made beyond the test set.
Multivariate Linear Regression	The model is similar to linear regression (v1), with the only difference that it undergoes certain modifications to have polynomial attributes. It is used to introduce polynomial characteristics in order to bring non-linearity into the scheme. Using the linear regression model on the transformed training data. It may use similar evaluation metrics as you would see in linear regression (v1). That is the same way as a model based on the test data points does not provide for predictions anymore.
ARIMA Model	Autoregressive Integrated Moving Average (ARIMA) is the model that is used. To make the time series stationary, this procedure normally includes differencing. Time series data after differencing might need an ARIMA model for identification. Usually, assessing performance would entail Mean Squared Error (MSE) or Akaike Information Criterion (AIC). The provided forecasts go beyond the test data to future time steps.

Continued on next page

Table 4.2 – Continued from previous page

Model	Description
Random Forest	An algorithm named Random Forest algorithm is used. For that, you might require scaling the features or encoding categorical features. This means tuning your data to train with a Random Forest model. Performance metrics such as Mean Squared Error (MSE) or Mean Absolute Error (MAE) are usually used in making this decision. This does not show predictions other than the test data.
LSTM (v1)	Making use of a Long Short-Term Memory (LSTM) network. Contains air traffic (windowing and scaling of time series data) radar data. Trains a model of long short-term memory modeling on windowed training data. Performance is assessed by criteria like Mean Squared Errors (MSE) or Mean Absolute Errors (MAE). Provides predictions for future time steps beyond the test data.
LSTM (v2)	Like LSTM (v1), this one too has additional capabilities such as dropout layers and scaling methods. It includes more features like dropout layers for regularization and different scaling techniques. Trains an LSTM model using different parameters that may increase the level of regularization. Performance is assessed using the same or similar standards as those employed by LSTM (V1) model. Predicts what will occur during the next period after the testing phase.

4.5.2 Observations

In the table 4.3 below we will give what observations we concluded for each model.

Table 4.3: Models and Observations

Model	Observations
Simple Linear Regression	1. Linear regression assumes a specific link between independent and dependent variables, but it may not be good enough if applied to stock market data, since some intricate patterns in such data fall outside its scope due to nonlinearity of the patterns. 2. In stock price forecasting, temporal relationships as well as long-term trends cannot be captured by it either.
Multivariate Linear Regression	1. Even with added polynomial feature transformation for capturing non-linear relationships in the stock prices, the new version might still be insufficient in capturing their complex dynamics. 2. Polynomial transformations can overfit data but not always because high degrees may sometimes lead to poor generalizations on some samples.

Continued on next page

Table 4.3 – Continued from previous page

Model	Observations
ARIMA Model	1. Some time series data might be difficult for ARIMA because they are non-stationary or irregularly spaced; however, such time series data could contain seasonality and/or correlation. 2. Stock prices usually behave nonlinearly and unstably which can hardly be adequately captured by ARIMA models without extensive preprocessing and differencing.
Random Forest	1. Using Random Forest make capturing complex data possible, but because they don't have a built-in sense of time, they're not good for time series analysis. 2. If the number of trees in the forest is too high it could lead to overfitting and can lead to poor generalization to new data.
LSTM (v1)	1. The first version of LSTM does not use regularization techniques such as dropout layers and this could lead to overfitting, especially when dealing with limited data. 2. It does not incorporate advanced scaling techniques or architectural optimizations that could improve its performance.
LSTM (v2)	1. LSTM (v2) addresses many of the limitations of the previous models: – It includes dropout layers for regularization, which helps prevent overfitting and improves generalization to unseen data. – The model architecture is optimized with additional features like different scaling techniques, which can enhance its ability to capture complex patterns in stock market data. – LSTM models inherently capture temporal dependencies and long-range patterns, making them well-suited for time series forecasting tasks like stock price prediction. – The ability to provide predictions for future time steps beyond the test data sets it apart from other models, making it more practical for real-world applications.

Summing up, each model has strengths and shortcomings but since it has the power to identify complex temporal structures, use regularization methods to avoid overfitting and generate forecasts for upcoming time periods, LSTM (v2) appears to be the preferable option for predicting stock prices.

4.6 Conclusion

In conclusion, Chapter 4 offers a comprehensive examination of stock price forecasting methodologies. Through meticulous data analysis and model implementation, we gain valuable insights into the dynamics of stock markets. While each forecasting model has its strengths and weaknesses, LSTM (second version) stands out for its ability to capture complex temporal patterns and provide accurate predictions for future time steps. These findings serve as a valuable resource for researchers and practitioners alike in the realm of financial forecasting.

Chapter 5

Realisation

5.1 Introduction

Our project realm is about channels using predictive models to analyse finance where the "realization stage" serves to mark the critical end of our venture by turning theoretical frameworks into concrete practice. Complete understanding of the practices which tied our progress together during those days is what this part is all about; a chapter that sheds more light on everything there is concerning actualization of our predictive stock analysis web app such as instruments and so on so forth, technologies or even libraries employed in it.

5.2 Work Environment

5.2.1 Introduction

The daily improvement of data science libraries and tools leaves us perplexed by the numerous ways that the same concept is implemented. This section will compare the various tools that we were thinking about utilizing and provide an explanation for our decisions.

5.2.2 Tools

In the upcoming section, we will list the softwares we decided to use. To complete our project, we had to use a variety of softwares to handle this task's different dimensions. The table 5.1 represents the tools we used

Table 5.1: Software Tools Description

Software	Description
	VsCode: 1 A free source-code editor developed by Microsoft; this editor supports numerous programming languages such as JAVA, CSS, HTML, and more. It includes extensions like TypeScript and HTML Tools, with built-in features to aid developers.

Continued on next page

Table 5.1 – Continued from previous page

Software	Description
 git	Git:2 Version control system used for tracking changes in source code during software development.
	GitHub:3 Web-based platform for hosting and collaborating on Git repositories, featuring issue tracking, pull requests, and project management functionalities.
	Jupyter Notebook:4 Interactive web application for creating and sharing documents with live code, equations, visualizations, and narrative text, commonly used for Python data exploration and analysis.
	Python:5 High-level programming language widely used in data science, machine learning, web development, and scientific computing.
	Node:6 JavaScript runtime environment for building scalable network applications, especially server-side applications.
	Postman:7 API development tool used for testing, debugging, and documenting APIs, enabling users to inspect response data and issue requests.

5.2.3 Technologies Used

This section highlights the technologies we applied in developing our project. Our coding process is usually fastened by employing many technologies. An extra tool can really be beneficial for the common tasks like testing, different scripts running and controlling the programming environment. The table 5.2 below represents the technologie we used in this project

Table 5.2: Technologies Used

Software	Description
	Python 3.12: Python programming language has made a major release in its 3.12.0 version and has a number of new features and optimizations.
	Git: Version control system is used to trace the modifications in source code while developing software.
	Angular:8 An online system that stores Git data and allows users to work on shared projects. It has many functions including issue tracking, pull requests, and project management.

Software	Description
	Html:9 An interactive website program that is used for building or disseminating articles that are embedded with real-time code snippets, equations, visualizations as well as plain text. It is mostly used for Python data inspection and computation.
	Css:10 Widely used programming language in data science, machine learning, web development, and scientific computing is known as AI (Artificial Intelligence).
	TypeScript:11 JavaScript runtime environment is utilized for developing scalable network applications including server-side applications.
 v 2.2.0	Clarity UI:12 This is an API development tool that is used for testing, debugging, and documenting APIs. Employing it enables those working with APIs to examine response data as well as issue requests in order to access these valuable resources.

5.2.4 Libraries

Here are the key Python libraries used in this project that accelerated coding tasks in various analyses, modeling and visualization scenario. These libraries are important because they contributed to increase productivity by enabling us to save time performing data-related tasks with essential functionality integration, thus leading to faster effective results through development simplicity. Below is the table 5.3 that represents the Libraries we implemented and used to make our project work .

Table 5.3: Technologies Used

Logiciel	Description
	Pandas:13 Pandas is a Python library that allows data manipulation and analysis easily. It has simple to use data structures such as DataFrame or Series, and it also provides data cleaning functions as well as transformation and visualization techniques.
	NumPy:14 In Python, NumPy is a fundamental package for numerical computation that comes with support for both arrays and matrices + mathematical functions that are efficient while operating on such.
	Scikit-learn:15 In Python, a machine learning library known as Scikit-learn offers tools to help with data pre-processing, features selection, model evaluation, and hyperparameter tuning.
	Tensorflow:16 It is a flexible solution built by Google for developing, training, and using deep neural network models in AI, unlike SAS, it's pedantically complex and requires large datasets.

Logiciel	Description
	Keras:17 Keras was created to be a user-friendly approach rather than a machine oriented one. Good cognitive design practices such as employing uniform and easy to use interfaces, reducing the number of user actions needed for common use cases while still allowing the software react to their mistakes are employed here. Additionally, it doesn't forget about providing excellent info available within its own project documentation or published guides.
	Matplotlib:18 With a variety of functions for creating static, interactive and animated displays, Matplotlib is a flexible charting software library meant to be used with the Python programming language.
	Seaborn:19 Seaborn is a statistical data visualization library that is built on top of Matplotlib and that can generate visually appealing graphs that are also informative.
	Plotly:20 A library which helps in creating high quality visualizations combining pictures and other interactive features is called Plotly.

5.3 Presentation

This section will showcase the general interfaces of our application, which ensure that the use case outlined is carried out.

5.3.1 Clustering Interface

Here is the Clustering Interface (figure 5.1) where we show the different cluster of the stocks in a graph and when sliding down we will find the List of Stocks

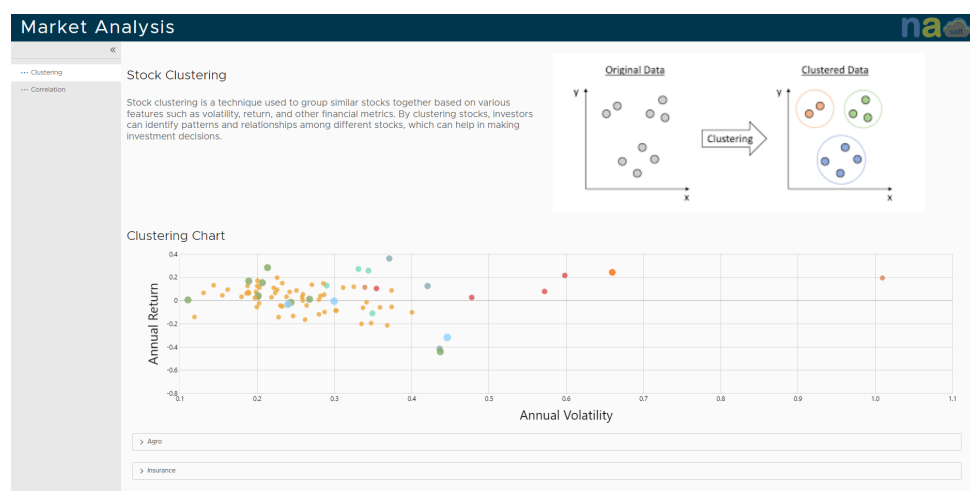


Figure 5.1: Clustering Interface

List of stocks

Here is the List of stocks (figure 5.2) where you can select the field and then choose the stock that you want to consult

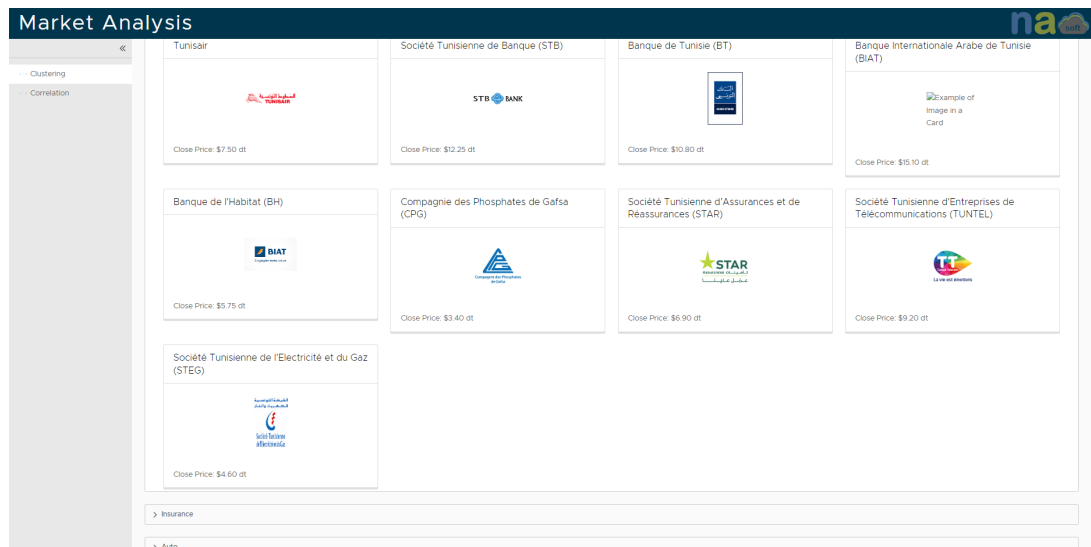


Figure 5.2: List of stocks

5.3.2 Correlation Interface

This interface (figure 5.3) displays the correlation between all the stocks

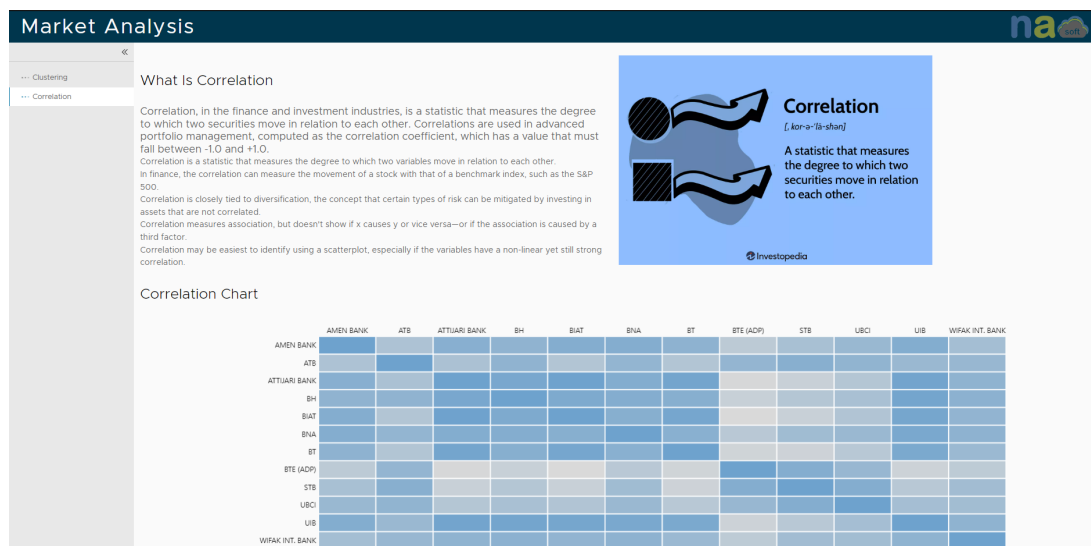


Figure 5.3: Correlation Interface

5.3.3 Single Stock Analytics Interface

This interface get displayed after clicking a single stock (figure 5.4) , it shows all the stock data (Adjusted Close , Adjusted High , Adjusted Low , Volume) on function of the date

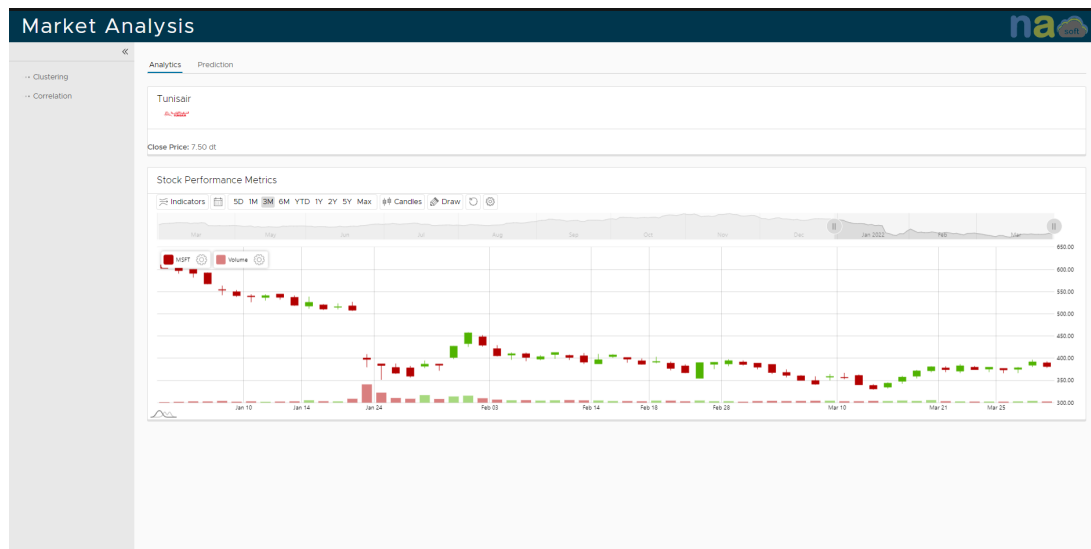


Figure 5.4: Single Stock Analytics Interface

5.3.4 Single Stock Prediction Interface

Here is the interface that will display the result of our model (figure 5.5) . The admin inputs how many future days he want to predict then the results will get shown on a graph (The results are the closing price)

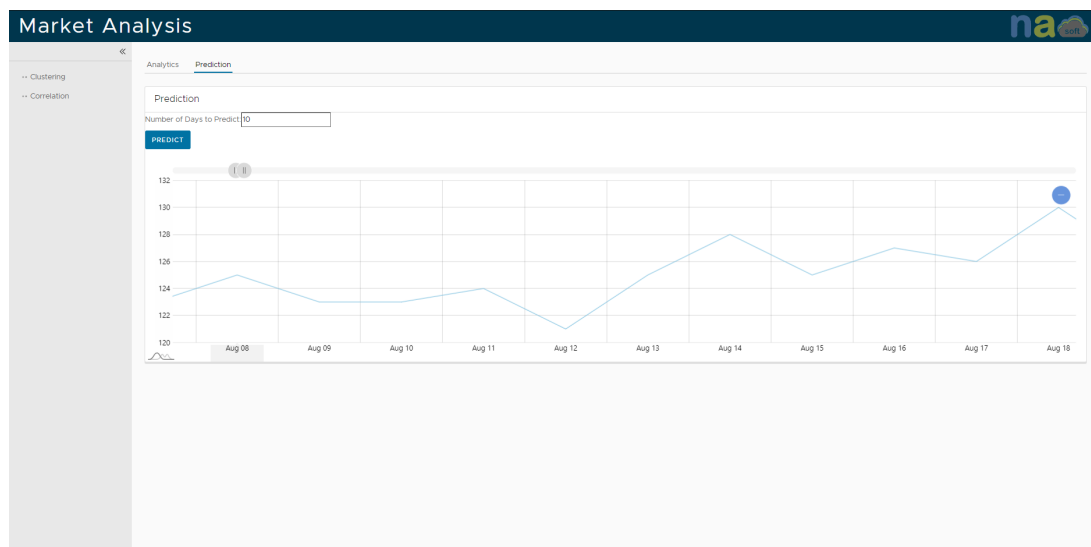


Figure 5.5: Single Stock Prediction Interface

5.4 Conclusion

This chapter has helped to elucidate how we implemented our online application for predicting stock market patterns, which is the final step in our project implementation. We have been able to make this application work so that it can respond to users' requests concerning anticipated analytics on financial markets of any sort.

General Conclusion

Finally, going through the complex aspect of predictive analysis as applied in the finance sector was fulfilling and eye-opening. We have covered a challenging yet promising journey from the start of our project to its practical realization. Consequently, we now know more about how technology interfaces with financial analyses.

We started by realizing that the modern financial markets complexities required some innovative solutions. After studying extensively, we found out how limited analytical methods in use today really are; this is when we discovered that there are better ways to predict future trends which could actually help us do something about them.

We researched the state of naive art, concerning the rationale behind prediction, learning several methods and tools. We looked closely at each of these approaches separately, from Long-Short-Term Memory to Random Forests, to decide which one would be appropriate for our initiative.

In the chapters that followed, we transformed theory into practice, kicked off a data exploration, analysis, and modeling process. With a strong methodology and plenty of information on hand, we managed to prepare data by using advanced analytical methods that helped us reveal important results and trends.

When we continued advancing we came across many challenges and took major decisions as well as fine-tuning our models and methodologies in order to improve the predictive accuracy and reliability. This has culminated in a comparison of the approaches that will help guide future endeavors. During this time frame, we have learned much about what works and doesn't work through detailed experimentation and evaluation.

In the realization stage, we gave life to our idea of converting theoretical ideas into real results. Using the most advanced technologies and protocols to do so, we designed a user-friendly web application that helps various parties to have workable data and make wise decisions.

Our trek has been one of exploration, creativity, as well as partnership after looking back. Having achieved what we set out to do in this chapter makes us feel proud and makes us look forward to the future journey. Our attempt is a proof of the great impact technology may have in changing financial analysis through forecasting techniques.

With the lessons and experiences we have gained, we move towards new horizons knowing that we can understand the complexities of the financial markets of tomorrow. As a group, we are prepared at the edge of innovation for what is to come and what it may require.

Netography

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- <https://code.visualstudio.com>
- Consulted at: 5/15/2024

2. Git

- <https://git-scm.com>
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Résumé

Ce rapport examine l'utilisation de méthodes de modélisation prévisionnelle avancées dans l'analyse financière pour relever les défis des marchés financiers complexes et riches en données. Le projet utilise des outils d'apprentissage automatique pour fournir des informations prédictives innovantes via une application web intuitive. L'objectif est d'améliorer la compréhension de la modélisation prédictive en finance pour aider à la prise de décision.

Mots clés: Methodologies avancées, Modélisation prévisionnelle, Analyse financière, Apprentissage automatique, application web

Abstract

This report investigates the use of advanced forecasting modeling methods in financial analysis to address the challenges posed by today's data-rich and complex financial markets. Traditional analytical approaches are often insufficient, necessitating innovative predictive insights. The project focuses on employing machine learning tools for predictive modeling and presenting the findings via a user-friendly web application designed for financial insights. Ultimately, it aims to enhance understanding of predictive modeling in finance through both theoretical discussions and real-world case studies, aiding market decision-making.

Keywords: Forecasting modeling, Financial analysis, Machine learning, Predictive modeling, Web application, Data analysis, Market decision-making